Department of Information and Communications Engineering

Machine Learning based GNSS Spoofing Detection and Mitigation for Cellular-Connected UAVs

Yongchao Dang

Aalto University

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Abstract

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Abstract

Cellular-connected Unmanned Aerial Vehicle (UAV) systems are a promising paradigm in the upcoming 5G-and-beyond mobile cellular networks by delivering numerous applications, such as the transportation of medicine, building inspection, and emergency communication. With the help of a cellular communication system and Global Navigation Satellite System (GNSS), UAVs can be deployed independently or collectively in remote and densely populated areas on demand. However, the civil GNSS service, especially GPS, is unencrypted and vulnerable to spoofing attacks, which threatens the security of remotely- and autonomously-controlled UAVs. Indeed, a GPS spoofer can use commercial Software-Defined Radio (SDR) tools to generate fake GPS signals and fool the UAV GPS receiver to calculate the wrong locations. Fortunately, the 3rd Generation Partnership Project (3GPP) has initiated a set of techniques and supports that enable mobile cellular networks to track and identify UAVs in order to enhance low-altitude airspace security.

The research works in this thesis leverage the potential of machine learning methods and 3GPP technique support to detect and mitigate GPS spoofing attacks for cellular-connected UAVs. The contributions of this thesis contain four parts. First, we propose a new adaptive trustable residence area algorithm to improve the conventional Mobile Positioning System (MPS) in terms of GPS spoofing detection accuracy under three base stations. Then, we deploy deep neural networks in the Multi-access Edge Computing (MEC) based 5G-assisted Unmanned Aerial System (UAS) for detecting GPS spoofing attacks, where the proposed deep learning methods can detect GPS spoofing attacks with only a single base station. Next, we analyze the max-min transmission rate for cellular-connected UAVs system theoretically and design an optimal Graphic Neural Network (GNN) to detect GPS spoofing attacks for cellular-connected UAV swarms. Finally, we employ a 3D radio map and particle filter to recover the UAV position and mitigate GPS spoofing attacks.

Keywords UAV, GNSS, GPS spoofing, machine learning

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Preface

This thesis was conducted at the Department of Communication and Networking, School of Electrical Engineering, Aalto University.

The research works in this thesis aim to leverage the potential of machine learning methods to detect and mitigate GPS spoofing for cellular-connected UAVs.

I would like to deeply thank Prof. Riku Jäntti for supervising this thesis and supporting me during the last two years of my Ph.D. journey. I really appreciate his patience and guidance on all matters related to the thesis.

I would like also to genuinely thank Prof. Tarik Taleb and Prof. Yulong Shen for supporting me throughout this road, without whom it would not have been possible to conduct and finish the majority of the research work in this thesis.

My gratitude also goes to Dr. Chafika Benzaid, Dr. Alp Karakoc, and Dr. Bin Yang for their comments and recommendations for improving the quality of this thesis.

I extend my thanks to my team members: Hamed Hellaoui, Ousama Bekkouche, Saba Norshahida and Azzam Al-Nahari.

A special thanks to the Chinese Scholarship Council funding and the Nokia Foundation for financial support.

Helsinki, August 11, 2023,

Yongchao Dang

Contents

List of Publications

This thesis consists of an overview and of the following publications which are referred to in the text by their Roman numerals.

I Y. Dang, C. Benzaïd, Y. Shen and T. Taleb. GPS spoofing detector with adaptive trustable residence area for cellular based-UAVs. In *2020 IEEE Global Communications Conference*, Taipei, Taiwan, pp. 1-6, Dec 2020.

II Y. Dang, C. Benzaïd, B. Yang and T. Taleb. Deep Learning for GPS Spoofing Detection in Cellular-Enabled UAV Systems. In *2021 International Conference on Networking and Network Applications (NaNA)*, Lijiang City, China, pp. 501-506, Dec 2021.

III Y. Dang, C. Benzaïd, T. Taleb, B. Yang, and Y. Shen. Transfer Learning based GPS Spoofing Detection for Cellular-Connected UAVs. In *2022 International Wireless Communications and Mobile Computing (IWCMC)*, Dubrovnik, Croatia, pp. 629-634, July 2022.

IV Y. Dang, C. Benzaïd, B. Yang, T. Taleb and Y. Shen. Deep Ensemble Learning Based GPS Spoofing Detection for Cellular-Connected UAVs. *IEEE Internet of Things Journal*, vol. 9, no. 24, pp. 25068-25085, Dec 2022.

V B. Yang, Y. Dang, T. Taleb, S. Shen and X. Jiang. Sum Rate and Max-Min Rate for Cellular-Enabled UAV Swarm Networks. *IEEE Transactions on Vehicular Technology*, vol. 72, no. 1, pp. 1073-1083, Jan. 2023.

VI Y. Dang, A. Karakoc and R. Jäntti. Graphic Neural Network based GPS Spoofing Detection for Cellular-Connected UAV Swarm. In *2023 IEEE 97th Vehicular Technology Conference*, Florence, Italy, June 2023.

VII Y. Dang, A. Karakoc, S. Norshahida, and R. Jantti. 3D Radio Map-based GPS spoofing Detection and Mitigation for Cellular-Connected

UAVs. Submitted to *IEEE Transactions on Machine Learning in Communications and Networking*, Nov 2022.

Author's Contribution

Publication I: "GPS spoofing detector with adaptive trustable residence area for cellular based-UAVs"

The author is the main contributor to this work and has implemented the solution as well as evaluated the results with support from the other authors. Dr. Benzaïd assisted in defining the problem and revised the paper, Prof. Shen advised on the main contributions of the paper and revised the manuscript, and Prof. Taleb supervised the process and reviewed the paper.

Publication II: "Deep Learning for GPS Spoofing Detection in Cellular-Enabled UAV Systems"

The author is the main contributor to this work and has edited the paper, designed and implemented deep learning neural networks for GPS spoofing detection in cellular-enabled UAV systems. Dr. Benzaïd advised on the main contribution of the paper and revised the paper. Dr. Bin assisted in problem formulation and system modeling. Prof. Taleb supervised the process and reviewed the paper.

Publication III: "Transfer Learning based GPS Spoofing Detection for Cellular-Connected UAVs"

The author is the main contributor to this work and has edited the paper, designed a convolutional neural network, and implemented transfer learning to the convolutional neural network for GPS spoofing detection. Dr. Benzaïd and Dr. Bin advised on the main contribution of the paper and revised the paper. Prof. Shen and Prof. Taleb supervised the process and reviewed the paper.

Publication IV: "Deep Ensemble Learning Based GPS Spoofing Detection for Cellular-Connected UAVs"

The author is the main contributor to this work, and has edited the paper, proposed the deep ensemble learning algorithm, implemented the solution, and evaluated the results. Dr. Benzaïd advised on the main contribution of the paper and revised the manuscript. Dr. Bin assisted in problem formulation and system modeling. Prof. Shen and Prof. Taleb supervised the process and reviewed the paper.

Publication V: "Sum Rate and Max-Min Rate for Cellular-Enabled UAV Swarm Networks"

The author is one of the main contributors to this work and has assisted in UAV swarm communication modeling, implemented the solution, and evaluated the results. Dr. Bin edited the paper and revised the manuscript. Professors Jiang, Shen and Taleb supervised the process and reviewed the paper.

Publication VI: "Graphic Neural Network based GPS Spoofing Detection for Cellular-Connected UAV Swarm"

The author is the main contributor to this work and has edited the paper, designed the graphic neural network, implemented the solution, and evaluated the results. Dr. Karakoc advised on the main contribution of the paper and revised the manuscript. Prof. Jäntti supervised the process and reviewed the paper.

Publication VII: "3D Radio Map-based GPS spoofing Detection and Mitigation for Cellular-Connected UAVs"

The author is the main contributor to this work and has edited the paper, built a 3D radio map using the Kriging method and ray tracing tools, implemented machine learning algorithms for spoofing detection and mitigation, and also evaluated the results. Dr. Karakoc advised on the main contribution of the paper and provided a 3D city map. Mrs. Norshahida revised the manuscript. Prof. Jäntti supervised the process and reviewed the paper.

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Abbreviations

3GPP 3rd Generation Partnership Project

A2A Air to Air

A2G Air to Ground

ATM Air Traffic Management

ATRA Adaptive Trustable Residence Area

AKF Adaptive Kalman Filter

AODV Ad-hoc On-demand Distance Vector

BS Base Station

CKF Cubature Kalman Filter

CNN Convolutional Neural Network

DT Decision Tree

EKF Extended Kalman Filter

FAA Federal Aviation Administration

FP False Positive

FN False Negative

FANET Flying Ad-hoc Network

GNSS Global Navigation Satellite System

GCNs Graph Convolutional Networks

GED Graph Edit Distance

GMLC Gateway Mobile Location Center

GNB Gaussian Naive Bayes

GNN Graphic Neural Network

GPS Global Positioning System

IMU Internal Measurement Units

INS Inertial Navigation System

KF Kalman Filter

LoS Line of Sight

LR Logistic Regression

LCS Location Services

MEC Multi-access Edge Computing

ML Machine Learning

MLP Multi-Layer Perceptron

MME Mobility Management Entity

MPS Mobile Positioning System

NGED Normalized Graph Edit Distance

NLoS Non-Line-of-Sight

NTN Neural Tensor Network

OSNMA Open Service Navigation Message Authentication

OCCS Operator Command and Control Service

PDF Probability Density Function

PGED Predicted Graph Edit Distance

RKF Robust Kalman Filter

RSRP Reference Signal Received Power

RSS Received Signal Strength

SDR Software-Defined Radio

SDPS Supplementary Data Provider Service

SVMG Support Vector Machine Gamma

SVMK Support Vector Machine Kernel

TESLA Timed Efficient Stream Loss tolerant Authentication

UAS Unmanned Aerial System

UAV Unmanned Aerial Vehicle

UK Universal Kriging

UKF Unscented Kalman Filter

UTM UAS Traffic Management

UTMS UAS Traffic Management Services

Symbols

Roman Symbols

d_E The tolerated GPS margin error

$\mathscr{F}$ A set of statistical results

G Group size

$\mathscr{G}_g$ The swarm GPS topology

$\mathscr{G}_l$ The swarm communication topology

I_0 The modified Bessel function

L Measured path loss

$\boldsymbol{L}$ Trust level

$\overline{L}$ Theoretical path loss

ΔL Path loss difference

$\mathscr{L}$ A group of path losses differences

$\mathscr{M}$ Radio map

N The number of particles

$\boldsymbol{P}$ GPS position

$\mathbb{P}_N$ The position set of N particles

$\mathfrak{r}$ Radio map RSS value

R BS's coverage range

$\boldsymbol{S_L}$ The trust region of $\boldsymbol{L}$

$\boldsymbol{Similarity}(\mathscr{G}_g, \mathscr{G}_l)$ The similarity between $\mathscr{G}_g$ and $\mathscr{G}_l$

$\mathscr{T}$ The adaptive trustable residence areas

$\boldsymbol{TL_R}$ The trust level of the reported GPS position

$\boldsymbol{TL_P}$ The trust level of the planned GPS position

u The target UAV

Greek Symbols

δ The trajectory deviation

φ A random angle

κ_φ The concentration of direction angle

μ_φ The mean of direction angle

$\boldsymbol{\omega}$ The weight

1. Introduction

This chapter first introduces the general background and motivation of this thesis to comprehend the readers of this work. In addition, it lists the objectives of the thesis and highlights the contribution of this work. The chapter also briefly summarizes the publications and shows the thesis structure at the end.

1.1 Context, Problem, and Objective of the Thesis

Unmanned Aerial Vehicles (UAVs), namely drones, are emerging as an integrated part of the Internet of Things (IoT) for delivering numerous applications, such as package transportation, infrastructure inspection, and emergency communication [2]. In addition, the upcoming 5G-and-beyond mobile networks promote remotely- and autonomously-controllable cellular-connected UAVs to market for their flexibility and mobility. According to the report in [3], the global UAV market is valued at nearly USD 26.2 Billion in 2022 and is expected to reach USD 38.3 Billion by 2027. However, the large number of UAVs introduces significant challenges to the current Air Traffic Management (ATM) system that is designed for human-controlled aircraft rather than for UAVs.

As a response, Unmanned Aircraft Systems (UAS) Traffic Management (UTM) systems are being developed by the Federal Aviation Administration (FAA) to manage air traffic and UAVs' and mission-related services for UAVs that include authentication, mission authorization, tracking and geofencing [4]. It is noteworthy that UTM systems highly rely on the Global Navigation Satellite System (GNSS) for air traffic management. Specifically, the Global Positioning System (GPS), owned by the United States of America, is officially used by UTM systems due to its global coverage and accuracy. Nevertheless, the civil GPS signals are unencrypted and vulnerable to spoofing attacks [5]. In practice, the GPS spoofing attacker can use Software-Defined Radio (SDR) tools to generate fake GPS signals and fool the UAV's GPS receiver with fake GPS positions [6].

Indeed, GPS spoofing can manipulate a UAV to bypass/breakthrough UTM regulations and mislead UAVs to no-fly zones, and/or give rise to collision risks. Therefore, appropriate measures must be incorporated into UTM systems to detect and mitigate GPS spoofing attacks.

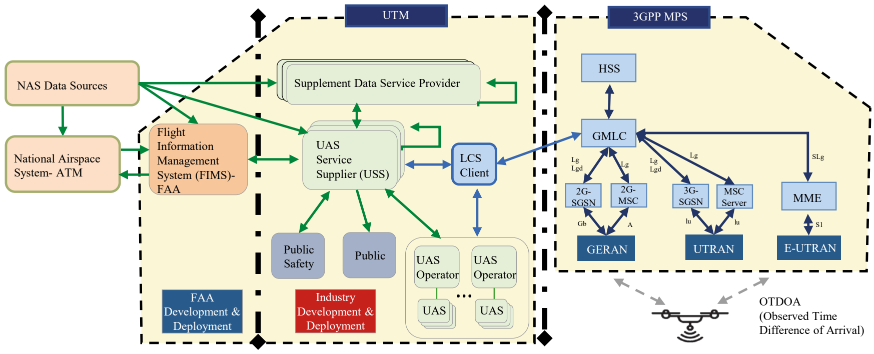


Figure 1.1. Connected UTM with 3GPP MPS [1]

The 3GPP Mobile Positioning System (MPS) can provide the location service to the UTM system for cellular-connected UAV air traffic management, shown in Figure 1.1. Rather than replacing the GPS, the MPS is an approximate positioning method from mobile networks that provide a simple and cheap way to detect GPS spoofing attacks for any cellular-connected UAVs by analyzing the GPS localization information in the telemetry data. As described in Figure 1.1, the UTM can access the MPS through a Gateway Mobile Location Center (GMLC) server from the 3GPP Location Services (LCS) system [1]. Correspondingly, the 3GPP Mobility Management Entity (MME) can also get the UAV flight plan approved by the UTM system to prepare cellular network configuration and facilitate handover procedures [1]. It is noteworthy that the handover procedure involves one or more cells and only one cell will serve the UAV, which depends on the Reference Signal Received Power (RSRP) measurement information from the cell-specific reference signals. In this way, the MPS can use the radio link measurements during the handover to verify the UAV GPS information. If the UAV is connected with a cell that has different GPS coordinates from the planned one, the GPS is considered to be spoofed. Once detected, the MPS will notify the UTM system and relocate the UAV position using the triangulation method and OTDOAs. At the same time, the UTM can set a dynamic no-flight zone and send alerts to the other UAVs and aircraft in the area via the 3GPP service [1]. Although the 3GPP MPS can help the UTM system detect GPS spoofing for cellular-connected UAVs, it only works during the cell handover and cannot indicate the UAV trajectory deviation inside the cell. Moreover, the triangulation method requires the UAV to be in an area covered by at least three Base Stations (BSs). Furthermore, the inherent highly-dynamic communication between the BS and UAVs introduces uncertain noise to the triangulation method, which decreases the accuracy of MPS-based UAV location.

The main objective of the thesis is to make GPS spoofing detection and mitigation far more efficient in 3GPP MPS by using advanced machine learning methods for cellular-connected UAVs. This efficiency enhancement entails a reduction in the localization error of MPS, an increase in the quantity of machine learning-based detection for UAV swarms, and a decrease in the time for training machine learning models. To overcome the aforementioned MPS weakness in terms of GPS spoofing detection and mitigation, we first need to improve the triangulation method that is used for GPS spoofing detection and mitigation during the handover. A better triangulation method cannot only increase the detection performance but also decrease the MPS location error. Secondly, it is important to endow a single BS with the ability to detect GPS spoofing when the UAV is inside the cell coverage area, which can ensure that the UAV mission is accomplished successfully in each cell. Indeed, the UAV keeps sharing its telemetry data with UTM through the 3GPP networks, and therefore the 3GPP MPS can obtain abundant signal measurements during the communication and analyze those data with machine learning methods to demonstrate the trajectory deviation caused by a GPS spoofing attack. Thirdly, since a flock of UAVs or a UAV swarm becomes popular in UAV applications for their advantages in economics and deployment, the continual monitoring and tracing of multiple UAVs introduce new challenges to the current 3GPP MPS. Similar to cellular-connected UAVs, all members in a UAV swarm are required to report their telemetry data to the UTM through an Air to Air (A2A) and Air to Ground (A2G) link, the kinds of wireless signal measurements which can also be employed to indicate abnormal behaviors in a UAV swarm. Fourthly, GPS spoofing mitigation is mandatory after the spoofing detection, because GPS spoofing mitigation can stop the alert sent by the UTM as well as help the UAV to find its real position. The challenge is that the triangulation method requires at least three BSs, which limits its usage to mitigate GPS spoofing when the UAV is located inside the cell. To counter such challenges, four main questions have been proposed in this thesis:

- How can MPS accuracy be improved in terms of GPS spoofing detection and mitigation during handover?
- How can GPS spoofing be detected for cellular-connected UAVs inside a cell?
- How can GPS spoofing be detected for cellular-connected UAV swarms?
- How can GPS spoofing be mitigated for cellular-connected UAVs and UAV swarms?

We have tackled the four main questions mentioned above through scientific articles, which are succinctly summarized in the following section. Furthermore, we have also delved into three basic questions pertaining to GNSS, GNSS spoofing, and anti-GNSS spoofing, as depicted in Figure 1.2.

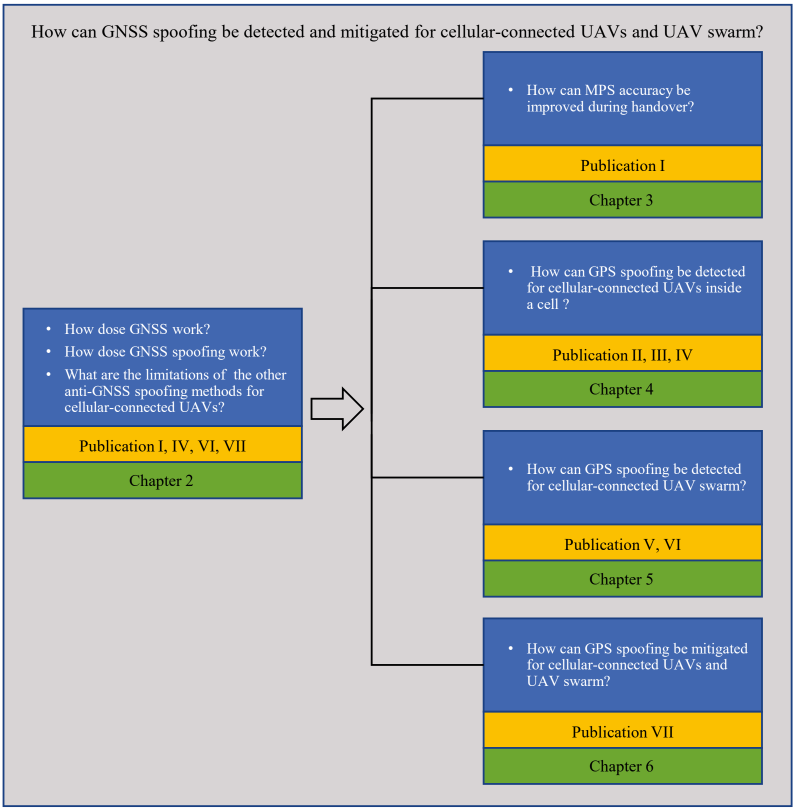


Figure 1.2. Research questions addressed in publications

1.2 Summary of Publications

This thesis is established in articles, where several solutions have been developed to address different research questions. Figure 1.2 shows all publications that correspond to the four main questions. Publication I investigates the cellular-connected UAV system and proposes a new area-based GPS spoofing detection method for the handover. Publications II, III, and IV use deep learning approaches to detect GPS spoofing when the UAV is inside the cell. Publications V and VI deal with cellular-connected UAV swarm problems. The GPS spoofing mitigation approach is given in Publication VII. The contributions of each individual publication are summarized as follows:

Publication I investigates a 5G-assisted UAV position monitoring and anti-GPS spoofed system, which allows lively verification of the UAVs' true positions and detection of GPS spoofing. In this publication, the Adaptive Trustable Residence Area (ATRA) is designed to estimate the UAV's real position by leveraging Received Signal Strength (RSS) measured by different BSs. The ATRA determines the GPS position non-spoofed when the UAV is located in the most trustable area while the other areas are given a different trust level with a possibility of spoofing.

Publication II uses a deep learning approach to enhance the MPS-based position monitoring and anti-GPS spoofing system for cellular-connected UAVs. Specifically, the Multi-Layer Perceptron (MLP) is used to decide the authenticity of the GPS positions and detect spoofing by analyzing the statistics features of the path loss from nearby BSs signals. The use of deep learning can detect GPS spoofing with only a single BS.

Publication III introduces the Convolutional Neural Network (CNN) and transfer learning in the edge servers to detect GPS spoofing for cellular-connected UAVs. The primary goal of this publication is to reduce detection latency and increase detection accuracy. Indeed, the use of transfer learning cannot only reduce the CNN training time but also increase the CNN detection accuracy, because it can transfer the CNN knowledge between edge servers when the UAV hands over from one BS to another.

Publication IV is an extension for publication II. It derives deep ensemble learning into edge servers to analyze the path loss statistical features and indicates the UAV trajectory deviation caused by GPS spoofing. Respectively, the MLP works on each edge server to extract the path loss features while the edge cloud server runs six different machine learning methods, including the MLP, Decision Trees (DTs), Logistic Regression (LR), Support Vector Machine Kernel (SVMK), Support Vector Machine Gamma (SVMG), and Gaussian Naive Bayes (GNB), to ensemble the individual edge server MLP results. Deep ensemble learning can continuously monitor UAV positions and detect GPS spoofing either during handover or inside the cell.

Publication V analyzes the sum rate maximization and max-min rate for a cellular-connected UAV swarm network. It formulates the communication model between the BS and UAV with two nonlinear optimization problems with the constraints of UAV transmission power and BS antenna parameters. Subsequently, the iterative algorithm is leveraged to solve the sum rate maximization problem, and the nonlinear Perron-Frobenius theory is applied to the max-min rate problem.

Publication VI deals with the issue of GPS spoofing attacks on cellular-connected UAV swarms. A Graphic Neural Network (GNN) is designed to detect GPS spoofing attacks on a UAV swarm by analyzing the similarity between the swarm's GPS topology and communications topology. In addition, the bipartite graph and Hungarian algorithm are used to build a simulator to test GNN performance. The GNN can detect GPS spoofing attacks on cellular-connected UAV swarms efficiently.

Publication VII aims to mitigate the GPS spoofing attack after detection. This paper leverage 3D radio map and machine learning methods to detect and mitigate GPS spoofing attacks for cellular-connected UAVs. Precisely, a 3D radio map is constructed with the help of ray tracing tools, deterministic channel models, and Kriging methods. Then, the MLP, CNN, and RNN are used to detect the spoofing while the particle filter is applied to relocate the UAV and mitigate GPS deviation using the 3D radio map and real-time RSS data. The mitigation can be deployed on the UAV as well as on the edge server.

1.3 Thesis Structure

The structure of the thesis is as follows. All chapters contribute to the question of how to detect and mitigate GPS spoofing attacks for cellular-connected UAVs and UAV swarms. Each chapter focuses on a question as shown in Figure 1.2.

Chapter 2 provides the literature background of UAV navigation systems and its security threats related to the topic of the thesis. This chapter first introduces the navigation system on a UAV, which includes the GNSS and the onboard Inertial Navigation System (INS). Regularly, the GNSS is used as the main navigation system by the UAV, and the INS works as a supplementary navigation system in order to improve the GNSS positioning accuracy. In addition, Chapter 2 also introduces the procedures of GNSS spoofing attacks on UAVs and the current INS-based anti-GNSS spoofing techniques.

Chapter 3 elaborates on the idea that aims to verify the UAV GPS position with the 3GPP MPS for the UTM. It proposes a 5G-assisted UAV position monitoring and anti-GPS spoofing system in order to address the problem of MPS accuracy during handover. This chapter also introduces the ATRA algorithm that is used to improve MPS accuracy. The contribution of this chapter is based on Publication I.

Chapter 4 investigates the simultaneous use of several cellular networks to verify the UAV GPS position with the 3GPP MPS for the UTM. It addresses the second question. This chapter introduces deep neural networks into edge servers for verifying the GPS position, which can simultaneously monitor the UAV positions and detect GPS spoofing when the UAV is located in an area covered by multi-BSs or even a single BS. The contribution of this chapter is based on Publications II, III and IV.

Chapter 5 focuses on GPS spoofing detection for a cellular-connected UAV swarm. This chapter addresses the third question. Since the cellular-connected UAV swarm has a different network topology compared with cellular-connected UAVs, an efficient GNN-based GPS spoofing detection approach has been introduced in this chapter, which can use the swarm topology information to detect GPS spoofing. The contribution of this chapter is based on Publications V and VI.

Chapter 6 focuses on GPS spoofing mitigation for cellular-connected UAVs. It addresses the last question. This chapter uses the 3D radio map and a machine learning method to relocate the UAV after spoofing detection. The contribution of this chapter is based on Publication VII.

Chapter 7 summarizes this thesis and discusses further research.

2. Background and System Frameworks

This chapter investigates UAV navigation systems and introduces GNSS spoofing and anti-GNSS spoofing techniques on UAVs. Regularly, the UAV uses both GNSS and INS to obtain its states, such as position, speed, acceleration, and orientation. Specifically, the INS is used as a supplementary navigation system to improve the GNSS position accuracy. However, GNSS spoofing attacks threaten the security of the UAV navigation system. Therefore, this chapter also introduces the procedures of GNSS spoofing attacks on UAVs and the literature on anti-GNSS spoofing techniques.

2.1 UAV Navigation System

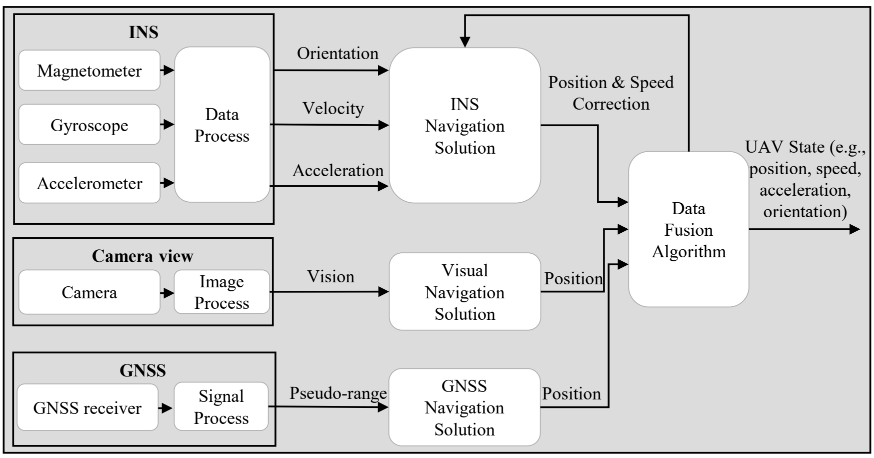


Figure 2.1. UAV navigation system

Figure 2.1 illustrates UAV navigation systems, which merge the INS, camera view, and GNSS with a data-fusion algorithm to produce the UAV state. Especially, the GNSS is the main navigation system that uses signal processing techniques to compute the pseudo-range between the UAV and navigation satellites and calculate the UAV position. In addition to GNSS, the INS uses Internal Measurement Units (IMUs), such as a magnetometer, gyroscope, and accelerometer to obtain the orientation, velocity, and acceleration of the UAV, which enhances GNSS accuracy and protects the GNSS from spoofing attacks. Furthermore, with the development of artificial intelligence, the camera view endows the UAV with bird vision for environment recognition and localization, which is a new solution for UAV navigation. The details of these systems are given in the following subsection.

2.1.1 GNSS

GNSS is a general concept of the system that uses a constellation of satellites to provide global positioning, navigating, and timing services to ground users. Examples of GNSS include the American GPS, the Russian GLONASS, the Chinese BeiDou and the European Galileo [7]. All of these systems work in the same manner that firstly estimates the signals propagation time from the in-view satellites to the ground user and then calculates the user position based on the pseudo-ranges of propagation time [8]. For instance, the American GPS, the most popular GNSS system around the world due to its global coverage and accuracy, uses 31 orbit satellites and the L1 frequency band (1575.42 MHz) to continuously broadcast navigation messages including time, the ephemeris data of satellites and the other information to ground users. Uniquely, the civil GPS navigation messages are encoded with the public coarse-acquisition (C/A) code and comprise 25 frames that consume 12.5 minutes to be transmitted [9]. Additionally, the satellites' ephemeris data is updated every 2 hours and is valid for 4 hours [9]. On the ground, the GPS receiver collects the raw GPS RF signals and decodes the GPS navigation messages, and then computes the pseudo-range using the timing information and ephemeris data. To calculate the receiver position, it requires at least four satellites in view of the GPS receiver. It is noteworthy that the GPS coordinate is located at a point but with uncertain errors due to the Doppler impacts and the hardware flaws in both GPS satellites and the receiver [10, 11]. Thus, the GPS receiver is typically combined with the INS to improve navigation performance.

2.1.2 Inertial Navigation System

The INS plays a crucial role in the UAV navigation system due to its capability to provide continuous, high-quality location data with low latency. [12]. As demonstrated in Figure 2.2, a typical INS uses different IMUs to measure the UAV current states based on the last one, including orientation, acceleration, velocity, and position. Fundamentally, the gyroscope measures the rate of angular rotation to ascertain the orientation of the

UAV, the magnetometer serves to indicate the UAV's heading, and the accelerometer records acceleration data, collectively providing velocity information for the UAV [13]. However, due to the imperfect of IMUs, the measurements of the state are with uncertain errors that are accumulating with navigation time [14]. Therefore, the combination of GNSS and INS can provide the UAV with effective and efficient navigation by leveraging the global coverage of GNSS and the continuation of INS [15].

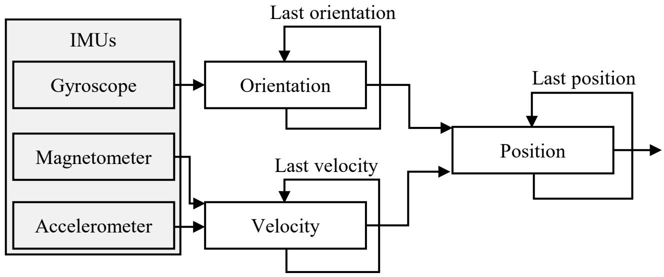


Figure 2.2. Diagram of INS

The Kalman Filter (KF) and its variations are persistently used to merge GNSS and INS for enhancing the performance of UAV navigation. Respectively, the KF is used for linear systems and its variations are for nonlinear systems (e.g., the Extended KF (EKF)) or both linear and nonlinear systems (e.g., the Unscented KF (UKF)) [16]. In [17], the authors proposed an Adaptive KF (AKF) that uses multiple weighted KFs to integrate the data from different sensors with GPS in order to fulfill the accuracy requirements of kinematic applications. The authors in [18] introduced a robust nonlinear filter to solve the initialization problem of the EKF in terms of UAV localization. Similarly, Wu *et al.* in [19] implemented an online constrained-optimization method to estimate the initialization errors and bias in KF-based INS/GNSS integration navigation applications. Subsequently, in [20], Hu *et al.* developed a direct filter approach against the kinematic model error in the UKF. Afterward, Hu *et al.* delivered a predictive model based on the UKF in [21] to eliminate these prediction errors. Furthermore, a fuzzy strong tracking Cubature KF (CKF) was designed by Chang *et al.* in [22] to identify and respond to the dynamics in the INS/GNSS integration. Moreover, the authors in [23] presented a hypothesis test-constrained Robust KF (RKF) to deal with the measurement outliers during the integration of INS and GNSS. In addition to the KF, [24] evaluated a factor graph optimizer that can integrate INS and GNSS for real-time positioning. Besides, neural networks are also emerging as new solutions to integrate INS and GNSS [25, 26]. Although the integration of INS and GNSS can achieve the highest precision and continuous navigation for vehicles, it is not a perfect navigation solution for UAVs that are controlled remotely [27]. On the one hand, commercial UAVs commonly use a strap-down INS integrated with an asynchronous

Kalman Filter, suitable for applications like aerial photography. However, cellular-connected UAVs, which are operated remotely for tasks, such as precision agriculture, search and rescue, and surveillance, demand higher positional accuracy. On the other hand, though highly accurate INS systems exist, their higher costs may lead operators to choose less expensive but potentially less accurate INS systems.

2.1.3 Camera View

The camera view is the foundation of computer vision that is used as a visual odometry for autonomous vehicle navigation [28]. When the camera view is deployed on an aircraft, it endows the UAV with a bird's ability to recognize the environment and locate itself for navigation [29]. In fact, the UAV camera view is not only used for infrastructure inspection but also for UAV navigation. As illustrated in Figure 2.3, the camera view employs motion analysis and image matching to produce the UAV's position. Specifically, motion analysis uses UAV images to obtain orientation information while image matching is applied on both UAV images and satellite maps to find the coordinate information. According to the diagram of camera view in Figure 2.3, the authors in [30] delivered a visual odometry-based UAV navigation approach, where the optic flow and stereo were used to compute the 3D position of a UAV. Thereafter, the authors in [31] proposed a deep CNN-based UAV autonomous navigation solution that can control the UAV path using only visual information. Similarly, Aldrich *et al.* introduced PoseNet CNN for UAV localization with an error of 2.5 meters [32]. Afterward, Samano *et al.* in [33] presented a vision-based geolocation method using image and map embedding methods. In addition to navigation, the UAV camera view can take aerial photos for updating the satellite map or building images map, which provides a practical closed-loop solution for UAV navigation [34, 35]. However, the camera view is vulnerable to weather changes. For example, clouds, rain, and fog can block the camera view and stop the UAV visual navigation system.

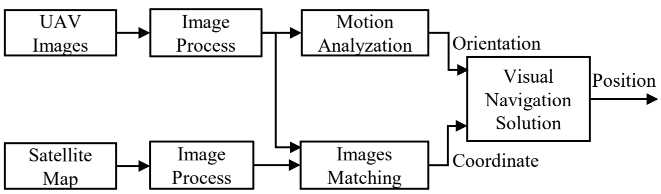


Figure 2.3. Diagram of camera view

2.2 GNSS Spoofing and Anti-GNSS Spoofing on UAVs

GNSS plays a key role in UAV navigation systems. However, civil GNSS messages are unencrypted and vulnerable to spoofing attacks, which threaten the security of the UAV navigation system. In practice, GNSS attackers can use inexpensive hardware to send fake GNSS signals or broadcast illegal GNSS messages in order to mislead UAVs to deviate from their planned trajectories. This section provides the procedures for GNSS spoofing on UAVs and anti-GNSS spoofing approaches given in the literature.

2.2.1 GNSS Spoofing on UAVs

GNSS is the main navigation solution for UAVs. A specific GNSS, named GPS, is widely equipped by UAVs and other civil applications for its global coverage and accuracy. However, the civil GPS waveform is designed following open documents and is predictable by using specifications in those documents [36]. Nonetheless, the civil GPS message is modulated on those waveforms without encryption and authentication. Consequently, one can easily spoof the civil GPS signals and messages with those predictable signal waveforms and public message structures [36].

There are two kinds of GPS spoofing attacks based on spoofed GPS signals generation methods [37]. One method generates a spoofed GPS signal with a relay while the other one uses a generator [37]. The former method collects the authentic GPS signals and then replays them with a certain time delay using higher transmission power, which leads to the same deviations in all pseudo-ranges calculation [37]. The latter method can counterfeit the GPS signals with the same structures as the authentic ones but with the spoofed information that set any positions and velocities by careful calculation [37]. Compared with the signal generator method, the relay method is cheaper and easier to implement but cannot set the spoofing position arbitrarily. Although the signal generator method can set the spoofed position and velocity to the target, it requires more expensive hardware as well as complex procedures, as shown in Figure 2.4.

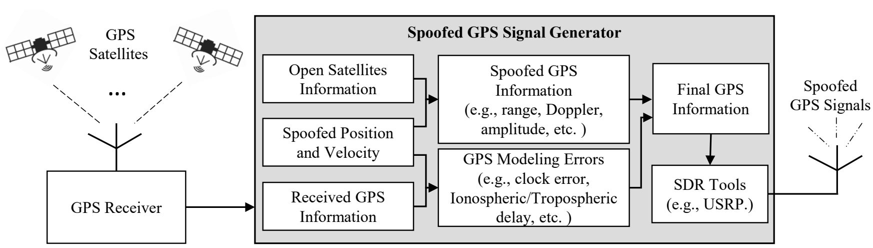


Figure 2.4. GPS spoofing with a signal generator

It is worth mentioning that the spoofed signal must be broadcast to the target position once receiving the authentic signals. Otherwise, the target GPS receiver will recognize the spoofed signal as the noise at the receiver if the delay is over the receiver tolerance [36]. In addition, the GPS signal generator has to download the real-time satellite information (e.g., satellite ephemeris, position and velocity, etc.) from the NASA website which publishes GPS satellite information, and then accordingly generates the GPS information with the spoofed position and velocity. Furthermore, the final GPS information includes both spoofed information as well as the modeling errors between the authentic GPS information and the counterfeit GPS information, including satellite clock error, ionospheric delay, and tropospheric delay [37]. Moreover, the SDR tools are in charge of modulating the final GPS information into the L1 band and sending the spoofed signal to the target position.

In recent years there has been a growing interest in the use of GPS spoofing techniques to capture and control UAVs that fly without permission. In [38], Kerns *et al.* theoretically analyzed UAV navigation system weakness and demonstrated GPS spoofing attacks that deviate a Hornet-MiniUAV from its planned path in practice. Following, Seo *et al.* in [36] developed a software-based GPS signal generator to capture the UAV using counterfeited GPS signals that are with the same structure but different navigation messages as the authentic ones. Then, the authors in [39] presented a free online software tool, named bladeGPS simulator, which can construct GPS messages the same as real satellites and spoof the UAV location. After that, in [5], the authors demonstrated a covert GPS spoofing algorithm to hijack the UAV by changing the planned trajectory slowly. Subsequently, Ceccato *et al.* proposed a spatial GPS spoofing method that uses multiple antennas and a Wiener filter to generate directional GPS spoofing signals towards a UAV swarm [40].

To overcome the weakness of civil GNSS, a lot of work has been done that aims to counter GNSS spoofing attacks, which includes designing a safer GNSS receiver, upgrading the current GNSS system, or improving INS performance in a GNSS-denied environment. Those works are summarized in the next part.

2.2.2 Anti-GNSS Spoofing on UAVs

Anti-GNSS spoofing methods can be broadly categorized into five classes, including navigation signal analysis approaches, cryptal-protected navigation message approaches, INS approaches, mobile cellular network approaches and machine learning approaches.

Navigation Signal Analysis Approaches

GNSS navigation signal analysis is an effective approach to detect spoofing attacks, which either uses multi-antennas techniques to analyze the DoAs of GNSS signals or leverages the cross-correlation between unencrypted civil signals and encrypted military signals to identify the spoofed signals. The spoofing detection approach in [41] used a multi-antenna technique to measure the DoA of GPS signals and distinguish spoofed signals from authentic signals. Compared with authentic GPS signals, spoofed signals are likely to have different DoAs. In reality, the authentic signals are from different GPS satellites in space but the spoofed signals are from the same source in the air. Similarly, the authors in [42] proposed a spatial signal processing approach that adopted multi-antennas reception and null-steering to identify and filter out fake GPS signals as well as mitigate the GPS spoofing. The aforementioned methods require a large-size multi-antenna that can measure the DoA of GPS signals. In addition to the multi-antenna technique methods, the works in [43] and [44] detected spoofed GPS signals by analyzing the cross-correlation between civil GPS signals received by a trusted receiver and military GPS signals provided by the defended receiver. GPS signals are considered spoofed when the cross-correlation is lower than a preset normal value. Although the adoption of the cross-correlation methods of signals requires no multi-antenna GPS receiver, they need cooperation between the trusted receiver and a defended receiver to conduct the cross-correlation analysis.

Cryptal-protected Navigation Message Approaches

Cryptography provides another available method that can prevent the civil GNSS receiver from spoofing attacks. In cryptography, encryption can ensure the confidentiality of navigation messages and assure the navigation message without malicious modification, while authentication can verify the integrity of navigation messages and ensure that the navigation messages are coming from the real GNSS satellites [45, 46]. The works in [47] encoded the digital signature into the GPS navigation messages and introduced a statistical hypothesis test to anti-GPS spoofing. The authors in [48] used a trusted execution environment to generate signatures for GPS navigation messages and prevented counterfeits. In addition to GPS navigation protection, Wu *et al.* in [49] leveraged the elliptic curve digital signature algorithm to sign the BeiDou-II navigation message, and then in [50] introduced the SM cryptographic algorithms to authenticate the BeiDou-II navigation messages. For the European Galileo, the Open Service Navigation Message Authentication (OSNMA) provides authentication services to all enabled receivers through the Timed Efficient Stream Loss-tolerant Authentication (TESLA) protocol [51]. Specifically, the OSNMA uses the TESLA chain key and TESLA root key to compute the tag and compares the computed tag with the received tag in the OSNMA data to verify the authenticity and integrity of the navigation message. Respectively, the TESLA root key includes a digital signature that is authenticated by a public key in the receiver, while the TESLA chain key is derived from a one-way hash chain and can be authenticated with a previous chain key or with the TESLA root key. Even though cryptography methods are practical to protect a GNSS receiver from spoofing attacks, they consume perceptible computing resources due to the key authentication and this induces appreciable latency for signature verification [52]. In addition, it is still a challenge for cryptography methods to withstand GNSS signal replay attacks [53].

Inertial Navigation System Approaches
INS is an alternative navigation solution that can either work independently using IMU sensors to produce a coarse location or collaborate with GPS to calculate an accurate position. In addition, INS is also available to cross-validate the authenticity of the GPS position. Khanafseh *et al.* designed a residual-based receiver that can autonomously monitor the GPS location with IMU sensors [54]. The works in [55] and [56] used the acceleration difference between IMU sensor measurements and GPS estimations to determine whether the GNSS has been hijacked or not. Then, in [57], the authors leveraged the onboard gyroscopes' measurements to detect GPS spoofing. In [58], Liang *et al.* integrated a rotational INS with GPS using an anti-spoofing KF. Similarly, Sathaye *et al.* leveraged EKF and IMU sensors to build a safe GPS receiver to detect and mitigate spoofing attacks [59]. The advantages of INS do not rely on any external signals and are immune to GPS spoofing. The disadvantage of INS is the IMU measurement error accumulation over time, which has negative impacts on spoofing detection.

Mobile Cellular Network Approaches
With the support from the 3GPP specification, a mobile cellular network is planned to provide localization and anti-GNSS spoofing services to IoT devices, for example, cellular-connected UAVs. In mobile cellular networks, the MPS has the ability to relocate the target and discriminate the spoofing attack. In [60], the authors exploited a 2G cellular network to relocate the vehicle position by exploiting the signal strength received from nearby BSs. The GPS is considered spoofed when the distance between the GPS position and the cellular network position is outside of a preset margin. The works in [61] proposed a new solution that checks the validity of the GPS position using the neighboring cells' information. The GPS position, claimed by the connected device, is spoofed if the device is not included in the cell coverage area. In addition, 5G massive MIMO is also available to track the UAV position through mobile cellular networks by exploiting the different spatial signatures of users [62]. Furthermore, the 3GPP offers technical support and standards to achieve precise positioning with low latency by utilizing dedicated 5G positioning reference signals [63, 64] as demonstrated in the 3GPP Release 18. Leveraging the resources provided

by 3GPP, this thesis employs machine learning techniques and 5G network infrastructure to detect and counteract GPS spoofing for cellular-connected UAVs.

Machine Learning Approaches

Machine learning belongs to the field of artificial intelligence that learns from historical data to unveil hidden patterns, which is recently a trend-driven approach for GPS spoofing detection. Machine learning-based spoofing detection approaches in cellular-connected UAV environments include GPS signal classification-based spoofing detection and GPS information verification-based spoofing detection.

Machine learning-based GPS signal classification methods take advantage of classification models to discriminate the spoofed signals from the actual GPS signals. The authors in [65] used the GPS signal features to train a neural network model and detected the spoofed signals from the raw ones, where the features consist of phase, energy, and imaginary components extracted by the tracking loop. Similarly, the work in [66] exploited the supervised neural networks to analyze the pseudo-range, Doppler shift, and signal-to-noise ratio of the received GPS signal for spoofing detection. In [67], Semanjsk *et al.* leveraged a Support Vector Machine (SVM) to detect GPS spoofing by analyzing the cross-correlation of GPS signals received from different GPS receivers. In [68], the authors applied different kinds of neural network models on satellite signal's radio frequency interference to detect GPS spoofing with the assumption that the spoofer cannot nullify satellite signals. It is noteworthy that the GPS signals are vulnerable to space weather changes that diminish the signal features [69]. In addition, the GPS jamming attack can broadcast the frequency noise and blind the GPS receiver, which considerably decreases the effectiveness of those approaches [70].

Machine learning-based GPS information verification methods make use of sensors-fusion location and external signals to prove the authenticity of the GPS locations and improve the detection accuracy. The sensors-fusion method, for example, the camera view, endows the UAV with the ability to recognize the surroundings and locate itself. In [71], the authors developed a hidden Markov model to analyze aerial photos taken by the UAV and detected the spoofed positions based on comparing the aerial photos with satellite photos. Using the PoseNet, a public convolutional neural network for image processing, the works in [32] took aerial images to prove the UAV GPS location and detect GPS spoofing. Even though the sensors-fusion techniques can relocate UAVs in a GPS spoofed or denied environment, the aforementioned camera-view-based spoofing detection methods spend perceptible time on aerial image processing and leads to noticeable detection latency. In addition to sensor-fusion methods, external signals, e.g., mobile cellular reference signals, can also prove the authenticity of GPS location in a cellular-connected UAV environment. Xiao *et al.* in [72] proposed the recurrent neural networks to detect GPS spoofing attacks on a UAV trajectory deviation, where the trajectory deviation is indicated by the DoA measurements from mobile cellular reference signals by the UAV. Nevertheless, the adoption of DoA measurements requires cylindrical antenna arrays on the UAV, which introduce more heavy payload and computation consumption on the UAV.

Table 2.1. Summary of detection methods for GPS spoofing attack

<table>
<tr><th rowspan="2">Detection method</th><th colspan="5">Requirements</th></tr>
<tr><th>Data source</th><th>Compute location</th><th>Hardware</th><th>Energy</th><th>Payload</th></tr>
<tr><td>Signal analysis</td><td rowspan="2">GPS signals</td><td rowspan="2">GPS receiver</td><td rowspan="2">Multi-receivers</td><td rowspan="2">UAV</td><td rowspan="2">Yes</td></tr>
<tr><td>ML-based signal analysis</td></tr>
<tr><td>Encryption</td><td>GPS message</td><td>GPS receiver</td><td>Secure hardware</td><td>UAV</td><td>No</td></tr>
<tr><td>Inertial sensors</td><td rowspan="2">Sensor data</td><td rowspan="2">UAV board</td><td rowspan="2">Different Sensors</td><td rowspan="2">UAV</td><td rowspan="2">Yes</td></tr>
<tr><td>ML-based GPS verification</td></tr>
<tr><td>Mobile cellular network</td><td>Cellular Signals</td><td>Edge Server</td><td>Multi-BSs</td><td>BS</td><td>No</td></tr>
</table>

Table.2.1 summarizes anti-GPS spoofing on UAV methods regarding the aforementioned related works. As already indicated, both signal analysis or ML-based signal analysis methods consume energy from the UAV to analyze the GPS signals and in some cases put more payload on the UAV to carry multi-receivers for proving either the direction of signals or a secure template. The encryption methods including OSNMA run at the GPS receiver secure hardware for GPS message encryption and decryption, which requires more energy but without extra payload on the UAV. The inertial sensors and ML-based GPS verification methods detect spoofing based on different kinds of sensor readings to prove the authenticity of the GPS position. However, these kinds of GPS information verification methods consume UAV energy for analyzing the sensor data and add more payload on the UAV for handling sensors. Although the mobile cellular network-based spoofing detection methods are deployed on the edge servers and without energy and payload demands on the UAV, they require that the UAV is located in the area covered by at least three BSs for conducting the triangular localization algorithms. Nevertheless, some airspace areas are covered by one BS, which reduces the performance of mobile cellular

networks tracking the UAV and detecting spoofing attacks.

Despite the merits of the aforementioned solutions in addressing the GPS spoofing issue in a cellular-connected UAV environment, a valid approach that can detect GPS spoofing effectively as well as adapt to energy consumption, resources limitation, and environmental constriction is still missing. To this end, we develop a set of new machine learning algorithms and leverage the advanced 5G-and-beyond mobile network facilities to devise effective GPS spoofing detection approaches for cellular-connected UAVs. The proposed approaches need no additional hardware or extra payload on the UAV and are impervious to environmental changes. In addition, by taking advantage of machine learning, the proposed approaches can detect GPS spoofing for either a single UAV or a UAV swarm with one BS. Moreover, the proposed approaches cannot only detect GPS spoofing attacks but also recover the UAV's real position by using a specific machine learning method and radio map techniques. Those works are summarized in the following chapter.

3. MPS-based GPS Spoofing Detection for a Cellular-Connected UAV

This chapter elaborates on the idea that uses the 3GPP MPS to verify the UAV GPS position for UTM and aims to improve MPS accuracy in terms of GPS spoofing detection during handover. Firstly, a 5G-assisted UAV position monitoring and anti-GPS spoofed system is designed according to the 3GPP LTE technique support in [73, 74, 75], which allows to verify of the UAV position and detect GPS spoofing. Then, the Adaptive Trustable Residence Area (ATRA) is proposed to estimate the UAV real position by leveraging the RSS measured by three BSs. The ATRA determines the GPS position non-spoofed when the UAV is located in the most trustable area while the other areas are given different trust levels with a possibility of spoofing. The contributions of this chapter are based on Publication I.

3.1 Problem Description

According to FAA regulation, UAVs are asked to send their GPS location and flight mission information to UTM for safety examination [76]. To support this regulation, the 3GPP announced a new technique that supports identifying and tracking UAVs over mobile networks [74]. As a response, Ericsson proposed architecture in [1] that allows tracking UAVs and detecting GPS spoofing through the 3GPP MPS service. However, the Ericsson system considers existing mobile cellular-network facilities rather than the 5G-and-beyond systems, and its detection accuracy depends on the cell coverage range. A big cell has a coarse location while a small cell gives a refined localization. Nevertheless, the conventional MPS is based on the triangular algorithm that locates the UAV at a position covered by three BS at the same time. The triangular location is inaccurate due to the uncertainty in signal measurements. Nonetheless, rather than a point, the real position is located in a residential area as illustrated in Figure 3.1. A residence area is highly dynamic for cellular-connected UAVs, which presents another challenge for MPS-based spoofing detection.

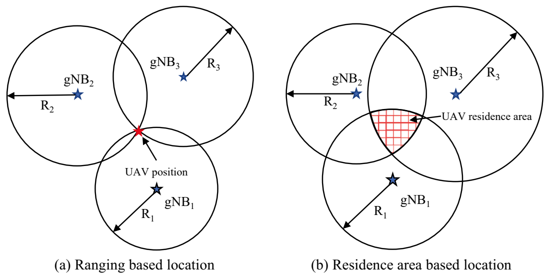


Figure 3.1. Triangular localization

3.2 5G-assisted UAV Position Monitoring and Anti-GPS Spoofing System

To address the aforementioned issues, a 5G-assisted UAV position monitoring, and anti-GPS spoofing system is proposed to verify the GPS locations of cellular-connected UAVs by leveraging up-link RSS measurements. In response to the low accuracy of MPS and the UAV's high dynamic flight, the ATRA method is developed to provide better localization and GPS spoofing detection for cellular-connected UAVs.

3.2.1 System High-Level Architecture

As illustrated in Figure 3.2, the proposed system uses a 5G network and facilities to provide communication and monitoring services for the UAV, UAV operator, and UTM. The system proceeds in five stages, including:

Stage 1. The UAV operator sends a flight mission request to the UTM through the 5G mobile network, specifying the mission type and its start/end GPS locations;

Stage 2. Once the request has been received, the UTM checks the mission's compliance with regulatory requirements, and the UTM issues the flight permission to the UAV if the mission defers to the regulations; otherwise, the UAV operator will be notified to change its mission plan;

Stage 3. Upon receiving the flight mission, the UAV must broadcast its GPS location in telemetry data periodically for collision avoidance and airspace enforcement [75];

Stage 4. Received by the nearby BSs, the telemetry data is augmented with the uplink RSS measurements and then forwarded to the UTM;

Stage 5. The augmented telemetry data is then processed in the UTM for estimating the UAV residence area with RSS measurements and checking the authenticity of the GPS position reported by the UAV; A warning message will be sent to the UAV operator if the UTM detects GPS spoofing.

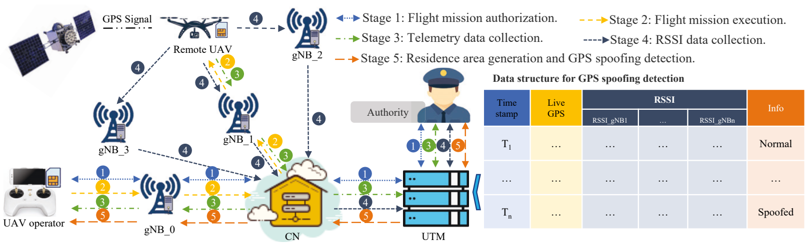


Figure 3.2. The proposed high-level architecture of the remote 5G UAV's position and GPS spoofing detector

The details on the residence area estimation and the GPS spoofing detection are provided in the subsequent sections.

3.2.2 Residence Area Estimation

To improve the conventional performance of MPS, we employ a residence area to denote the UAV's location. This residence area is determined using a trilateration method that relies on the distances calculated from the RSS measurements of three base stations, as depicted in Figure 3.1. However, the original size of the residence area is too large to validate the UAV's GPS authenticity effectively. To further refine the residence area and improve the spoofing detection performance, we divide the residence area into three sections, with each section assigned with different trust levels $L1$, $L2$, and $L3$, as shown in Figure 3.3.

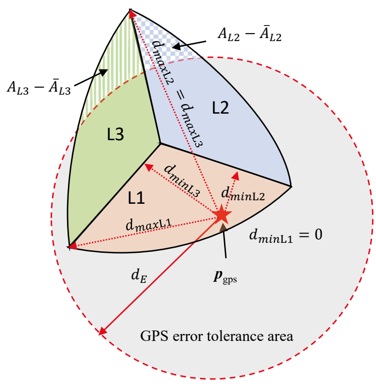


Figure 3.3. Adaptive Trustable Residence Area (ATRA)

Mathematically, the $L1$ area is defined as:

$$\begin{cases} d_1^2 = R_1^2 - [(x - X_1)^2 + (y - Y_1)^2] > 0 \\ d_2^2 = R_2^2 - [(x - X_2)^2 + (y - Y_2)^2] > 0 \\ d_3^2 = R_3^2 - [(x - X_3)^2 + (y - Y_3)^2] > 0 \\ d_1 \leqslant d_2 \wedge d_1 < d_3 \end{cases} \tag{3.1}$$

where (X_1, Y_1), (X_2, Y_2) and (X_3, Y_3) represent the location of the nearby BSs gNB_1, gNB_2, and gNB_3, and R_1, R_2 and R_3 denote the estimated distances between the UAV and BSs by using RSSI measurements. d_1 represents the maximum distance from the center of the residence area to the border of the L_1 area, while d_2 and d_3 denote the maximum distances to the borders of the L_2 and L_3 areas, respectively, as illustrated in Figure 3.4. The symbol '$\wedge$' in Eq. (3.1) is the logical AND operator. It represents the condition for the UAV to be within the residence area. (x, y) is a position inside the residence area. To obtain the $L2$ and $L3$ trusted area, the fourth line in Eq. (3.1) should be replaced by:

$$d_2 \leqslant d_3 \wedge d_2 < d_1 \tag{3.2}$$

$$d_3 \leqslant d_1 \wedge d_3 < d_2 \tag{3.3}$$

where Eq. (3.2) is for the L2 area and Eq. (3.3) is for the L3 area. The determination of the ATRA is summarized in Algorithm.1.

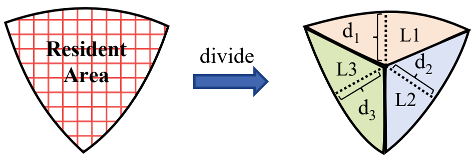


Figure 3.4. The residence area is divided into trust-level areas.

In Algorithm 1, the weight $\boldsymbol{\omega_i}$ is assigned to Li as a sign of trust level and is computed as:

$$\boldsymbol{\omega_i} = \frac{R_{k1} R_{k2}}{\sum_{l=1}^{2} \sum_{k=l+1}^{3} R_l R_k}, k1 \neq k2 \neq i \, (k1, k2 = [1, 2, 3]) \tag{3.4}$$

$\sum_{i=1}^{3} \boldsymbol{\omega_i} = 1$, and a smaller estimated distance is signed with higher trust. For example, the estimated distances R_1, R_2, and R_3 are $88m$, $136m$, and $170m$, and the corresponding trust weights are 0.46, 0.30, and 0.24, respectively. As depicted in Figure 3.3, the ATRA comprises either the entirety or parts of trust areas, represented by $\mathscr{T}$, each maintaining the same trust level as its original area. A trust area of level L_i is incorporated into the ATRA if the whole area falls within the GPS error tolerance area, or if the ratio of the non-overlapping area to the entire area is less than a threshold

Algorithm 1 ATRA Determination

Input:

$\boldsymbol{S_{L_i}}$: The Li trust region, $i = \overline{1,3}$
$\boldsymbol{\omega_i}$: The weight of the trust level Li
$\boldsymbol{P_{gps}} = (P_{latitude}, P_{longitude})$: The GPS position planned by the UTM
$\boldsymbol{d_E}$: The tolerated GPS margin error
$\boldsymbol{R_j}$: The RSSI-based estimated distance between the UAV and the base station j

Output:

$\mathscr{T}$: The adaptive trustable residence area

$(x_{gps}, y_{gps}) \leftarrow Get\_Cartisian(P_{gps})$
$\boldsymbol{T} \leftarrow \{\}$
for each L_i **do**
  $d_{minPi} \leftarrow Min\_Distance(S_{L_i}, (x_{gps}, y_{gps}))$
  $d_{maxPi} \leftarrow Max\_Distance(S_{L_i}, (x_{gps}, y_{gps}))$
  if $d_{maxPi} \le \boldsymbol{d_E}$ **then**
    $\mathscr{T} \leftarrow \boldsymbol{T} \cup \{\boldsymbol{L_i}\}$
  else if $d_{minPi} < \boldsymbol{d_E}$ and $d_{maxPi} > \boldsymbol{d_E}$ **then**
    $\boldsymbol{A_{L_i}} \leftarrow Area(\boldsymbol{S_{L_i}})$
    $\boldsymbol{\overline{A}_{L_i}} \leftarrow Area(\boldsymbol{S_{L_i}} \cap Circle(x_{gps}, y_{gps}, d_E))$
    if $\frac{A_{L_i} - \overline{A}_{L_i}}{A_{L_i}} < \boldsymbol{\omega_i}$ **then**
      $\mathscr{T} \leftarrow \boldsymbol{T} \cup \{\boldsymbol{L_i}\}$
    end if
  end if
end for
Return $\mathscr{T}$

determined by the weight $\boldsymbol{\omega_i}$. For instance, if the weights of L_2 and L_3 are 0.30 and 0.24 respectively, and the ratios of the non-overlapping areas to their respective trust areas are 0.28 and 0.30, then the L_2 trust area will be included in ATRA, but not the L_3 trust area.

3.2.3 GPS Spoofing Detection

GPS spoofing detection is executed in the UTM that compares the UAV-reported GPS location with the ATRA while accounting for GPS error, as seen in Algorithm.2. Specifically, the GPS position is assigned $L0$ when it is outside the ATRA. Under other conditions, the GPS position is assigned one of the trust levels $L1$, $L2$, and $L3$, depending on where the trusted area is located. The UAV-reported GPS position is considered as not spoofed if the trust level of the reported GPS position belongs to the ATRA or if both the reported and the whole ATRA are inside the GPS error tolerance area. The GPS is also not spoofing when the GPS error tolerance area is a part of the residence area and both the planned and reported GPS positions have the same trust level. Otherwise, the reported GPS position is considered spoofed.

Algorithm 2 ATRA-based GPS spoofing detection

Input:
- $\mathscr{T}$: The adaptive trustable residence area
- $\boldsymbol{TL_{R_{gps}}}$: The trust level of the reported GPS position
- $\boldsymbol{TL_{P_{gps}}}$: The trust level of the planned GPS position
- $\boldsymbol{d_E}$: The tolerated GPS margin error

Output:
- $\boldsymbol{Decision}$: The reported GPS position is spoofed (= 1) or not (= 0)

```
if (TL_{R_gps} ∈ T) then
    Decision ← 0
else if (card(T) == 3) and (distance(R_gps, P_gps) < d_E) then
    Decision ← 0
else if (TL_{R_gps} == TL_{P_gps}) and (TL_{R_gps} ≠ L_0) then
    Decision ← 0
else
    Decision ← 1
end if
return Decision
```

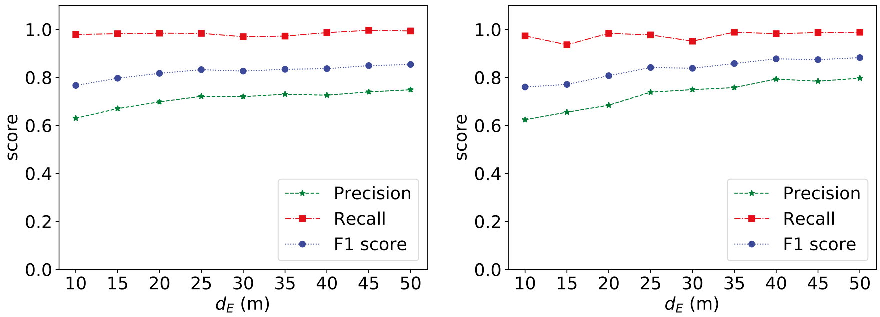
3.3 Performance Evaluation

The evaluation of an ATRA-based GPS spoofing detector is performed in both free space and urban space by measuring the detection *precision*, *recall* and *F1-score*, defined in Eq. (3.5).

$$\begin{cases} Precision = TP/(TP+FP) \\ Recall = TP/(TP+FN) \\ F1 = \dfrac{2 * Recall * Precision}{Recall + Precision} \end{cases} \tag{3.5}$$

where TP is the true positive that shows correct detection of spoofed positions, FN stands for the false negative indicating spoofed positions detected as normal positions, FP presents the false positive for the normal positions considered as spoofed, and TN is the abbreviation of true negative showing the normal positions correctly detected as not spoofed.

The simulation results are summarized in Figure 3.5, where the path loss exponent for free space and urban environment is randomly varied in [1.9,2.1] and [2.7,3.5] and the random shadowing effects is $6dBm$ for free space while $10dBm$ is set for urban space. There are three key observations from the obtained results. Firstly, the ATRA approach shows good performance in both free-space and urban area environments, which demonstrates the robustness of the ATRA against the shadowing effects. Secondly, the detection performance increases as we turn up the GPS error toleration and can reach 80% precision and 95% recall rate for a tolerated deviation of $50m$. Thirdly, the ATRA can achieve a balance between precision and recall as the overall F1-score values above 88%. It is worth noting that the performance between free space and urban areas depicted in Figure 3.5 shows little difference. This result can be attributed to the effectiveness of the ATRA method, which successfully compensates for the additional path loss and interference typically encountered in urban environments. Moreover, the shadowing effect has been modeled to closely match the characteristics of each environment, further reducing the disparity in performance between these two scenarios.



(a) Free space (b) Urban space

Figure 3.5. ATRA performance in free and urban space

Compared to "Drive Me Not", a GPS spoofing detection method via 2G cellular networks, presented in [60], the proposed ATRA significantly enhances detection accuracy and reduces detection latency for cellular-connected UAVs. Unlike "Drive Me Not" which uses downlink RSS and requires 150 seconds to detect anomalies within a 78-meter range, ATRA uses uplink RSS, is deployed on edge servers, and can detect spoofing within 50 meters with over 80% accuracy. Further, by integrating with 5G URLLC within the detection framework, ATRA enables real-time detection in one second, significantly outpacing "Drive Me Not".

3.4 Summary and Discussions

In this chapter, we introduced the 5G-assisted UAV position and anti-GPS spoofing system. The proposed system leverages the uplink RSS measurements collected by the nearby BSs to identify the UAV-reported position, where the ATRA is used to represent the region for non-spoofed GPS locations. The evaluation of ATRA shows the high effectiveness of GPS spoofing detection in both free-space and urban areas. Nevertheless, a key limitation of ATRA is its requirement for data from at least three base stations simultaneously, which may restrict its use in scenarios where the UAV is situated within a single cell. In the following chapter, we will explore the use of deep learning techniques for detecting GPS spoofing attacks on cellular-connected UAVs. This approach promises to offer continuous monitoring and validation of GPS positions on edge servers, regardless of the UAV's location and the number of BSs.

Compared with the state-of-the-art MPS-based methods, the proposed ATRA method is deployed on the edge server and provides different trust levels to the area where the UAV is located and therefore detects GPS spoofing with lower latency as well as a higher accuracy. The use of the ATRA-based GPS spoofing detection method requires RSS measurements from different BSs, which are obtained from statistical path loss models in the simulator without considering the real-time noise. For a better evaluation, path loss models should be extended in order to comprehend the real-time RSS estimation errors. Results have shown that setting bigger GPS errors in ATRA-based GPS spoofing detection can offer better performance in both free space and urban space. This approach is supported by [74]. It was agreed that the 3GPP mobile network needs to provide the list of authorized UAV information (e.g., location and identification) in the target area according to the UTM requests. In addition to the metrics evaluated in this thesis, the time required for detection, as well as the time interval between false alarms and the distance deviated from the reference path, have not been considered. This is because, in the detection system, the time needed to detect spoofing is influenced by an array of factors such as network latency, UAV speed, and the data transmission rate between the UAV and the base station. Despite not considering these metrics, the ATRA method proposed in this thesis stands as a viable and robust solution for UTM systems to effectively track UAVs through a 3GPP mobile network and to reliably detect GPS spoofing. In the future, we plan to incorporate detection time as a key performance indicator for the system.

4. Deep Learning-based GPS Spoofing Detection for Cellular-Connected UAVs

This chapter introduces deep neural networks into edge servers for verifying the GPS position, which can consecutively monitor the UAV positions and detect GPS spoofing when the UAV is located in the area covered by multiple BSs or even a single BS. Firstly, the Multi-Layer Perceptron (MLP) is used to analyze the statistics features of path loss from nearby BSs signals and decide the authenticity of the GPS positions reported by the UAV for the UTM. Specifically, the deep neural network can detect GPS spoofing with only one BS. Then, the Convolutional Neural Networks (CNN) are deployed on edge servers to monitor the UAV position and detect GPS spoofing. In addition, transfer learning is utilized to train the CNN models, which not only reduces the CNN training time but also increases the CNN detection accuracy, because it can transfer the CNN knowledge between edge servers from one BS to another. Finally, deep ensemble learning is proposed to ensemble the individual edge server MLP results. Deep ensemble learning can continuously monitor UAV positions and detect GPS spoofing either during handover or inside the cell. This chapter addresses the second question with three Publications (Publication II, III, and IV).

4.1 Problem Description

3GPP MPS service provides an affordable GPS spoofing detection solution for cellular-connected UAVs. Although the proposed ATRA-based GPS spoofing detection method accommodates resource, cost, and environmental constraints, it is only working on the handover area that is covered by at least three BSs. To detect GPS spoofing consecutively in both the handover area and inside-cell area where only one BS is visible, an effective and efficient GPS spoofing detector is required to overcome the weakness of the proposed ATRA method for cellular-connected UAVs. Therefore, machine learning methods come to mind that are able to learn from historical path loss data and unveil GPS spoofing patterns. Three main issues related to machine learning methods are addressed in this chapter:

1. First, the machine learning model uses path losses to indicate the UAV trajectory deviation caused by GPS spoofing attacks, but path losses are less prone to changes in a highly dynamic cellular-connected UAV environment. Therefore, the data from these path losses demand preprocessing to mitigate the environmental impacts in such a way that can improve the reliability of machine learning detection results.

2. Second, although the use of machine learning is a recent trend for GPS spoofing detection in cellular-connected UAV environments, the performance of machine learning models heavily depends on the volume of the training data; the more data are available and the more accurate machine learning model will be [77]. Therefore, the data needs to be shared among BSs so that a more accurate model can be trained as well as a higher GPS spoofing detection accuracy will be achieved.

3. Third, machine learning models are individually deployed on each BS to detect GPS spoofing in the inside-cell area, and, even in the handover area, these models work separately. Therefore, an ensemble method is required to merge these separate detection results into a final result with the purpose of improving the detection accuracy.

There are two kinds of machine learning methods available for GPS spoofing detection, including the MLP and CNN. the MLP requires a set of training data that are prepossessed with statistical methods in order to make the detection insensitive to environment changes, whereas the CNN can work on the raw data directly because its convolutional layers can help to extract the deep trajectory features directly from the original data. However, both the MLP and CNN are deep neural networks and require a large amount of training data to build an accurate detection model. Nevertheless, the adoption of path loss statistics drops some of the original features in the inputs and leads to a mistake in the detection results. Furthermore, it takes too much time and bandwidth to share those data among BSs, which may cause network congestion and degradation in network performance. Moreover, the conventional ensemble methods improve the detection accuracy by combining multiple results with the weighted hypothesis testing, which may not be efficient for cellular-connected UAVs in dynamic environments that require appropriate thresholds for hypothesis testing.

The remainder of this chapter is organized as follows. Section 4.2 presents the system modeling and proposed solutions. Section 4.3 evaluates the proposed solutions. Concluding remarks and discussions are presented in Section 4.4.

4.2 Modeling and Proposed Solutions

In this section, an MEC-based 5G-assisted UAS platform is first introduced. Then, the problem of GPS spoofing attacks on cellular-connected UAVs is reviewed and formulated mathematically. After that, deep neural networks are presented to detect GPS spoofing for cellular-connected UAVs.

4.2.1 System Framework

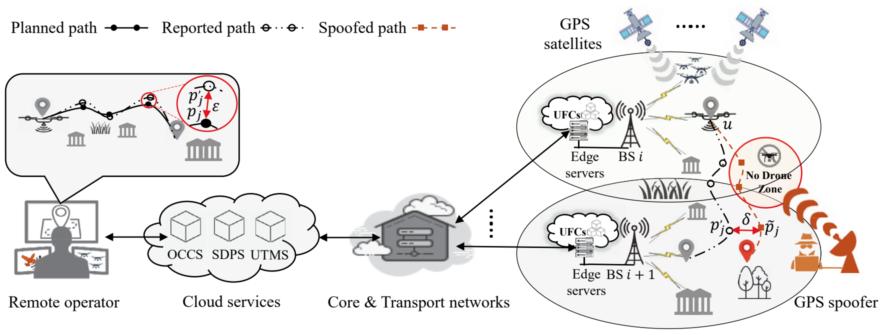


Figure 4.1. An MEC-based 5G-assisted unmanned aerial system

Figure. 4.1 shows the MEC-based 5G-assisted UAS architecture, which includes a remote operator, cloud services, core and transport networks, and cellular networks [78]. The remote operator can control and monitor remote UAVs through the Operator Command and Control Service (OCCS) in the cloud. Other cloud services, such as the Supplementary Data Provider Service (SDPS) and the UAS Traffic Management Services (UTMS), provide necessary meteorological data and airspace restrictions to make sure the mission is in compliance with regulations [79]. The core and transport networks bridge the connections between cellular networks and cloud services. In this system, the edge servers host the UFC services that monitor and control the UAV during flight missions. The advantage of this system is without additional hardware and energy requirements on the UAV.

4.2.2 Problem Formulation

As shown in Figure 4.1, the GPS spoofing scenario consists of a GPS spoofer, a victim UAV u, and BSs. The GPS spoofer broadcasts the fake signals on L2 bandwidths to the victim UAV. Respectively, BS i is located at (x_i, y_i, h_i) and BS $i+1$ is at $(x_{i+1}, y_{i+1}, h_{i+1})$. Without considering GPS spoofing and GPS error, the UAV u should be at the planned waypoint p_j at time t. Including GPS errors, at time t, the reported location is at $p_j^{'}$ and is far away from p_j with an error ϵ. When the GPS is spoofed at time t, the UAV is located at $\tilde{p}_j$ with a deviation δ from the planned position p_j, $\epsilon \leq dE < \delta$, and dE is the GPS margin error. However, the proposed ATRA may not detect this because the UAV is located inside the cell. Nevertheless, the testimony of the GPS spoofing attack is actually not vanished but hides in the path loss changes. Those changes can indicate a GPS spoofing attack, which is based on the fact that path losses are highly dependent on the distance between BS and UAV. By comparing the theoretical path losses calculated from the reported GPS position with the real path losses measured from the signal, the deviation caused by the GPS spoofing can be detected.

In this section, the theoretical path loss calculation follows 3GPP specifications in [80] which consists of both Line-of-Sight (LoS) links and Non-Line-of-Sight (NLoS) links. Let $\overline{L}_{iu}$ and L_{iu} denote the theoretical and real path loss between BS i and UAV u, respectively. ΔL_{iu} shows the absolute difference between $\overline{L}_{iu}$ and L_{iu},$\Delta L_{iu} = \left|\overline{L}_{iu} - L_{iu}\right|$. Hence, on BS i, the GPS spoofing detection problem can be formulated with a hypothesis testing in Eq. (4.1).

$$\begin{cases} H_0: & \Delta L_{iu} > T, \\ H_1: & \Delta L_{iu} \leq T, \end{cases} \tag{4.1}$$

where H_0 stands for the GPS position spoofed when ΔL_{iu} is above the threshold T, while H_1 represents no GPS spoofing. The detection of spoofing relies on the comparison of theoretical and real path losses. When the similarity between these two is close to zero, it implies GPS spoofing, making it logically consistent to label 'spoofing present' as H0 in the hypothesis testing. It is noteworthy that the real path losses L_{iu} are vulnerable to environmental changes, such as temperature, humidity, cloud, and fog. Thus, the GPS spoofing detection method in Eq. (4.1) may have failed in assessing the distance deviation caused by the GPS spoofing attack. In addition, the detection performance depends on threshold setting, a smaller threshold results in a higher false alarm while a bigger threshold leads to a higher probability of missed detection.

To overcome these problems, we leverage the potential of deep learning approaches to devise an effective GPS spoofing detection approach for cellular-connected UAVs in the following section, including the statistical MLP, CNN, transfer learning, and deep ensemble learning approaches for GPS spoofing detection.

4.2.3 Deep Learning Approach

In this part, we apply statistical methods and an MLP model on path loss data for GPS spoofing detection inside the cell. Then, the CNN is introduced to detect GPS spoofing without statistical processing, which can keep the original features in the path losses data and therefore improve accuracy as well as decrease latency.

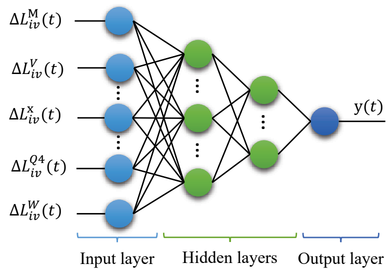


Figure 4.2. The structure of the MLP

Figure 4.2 demonstrates the proposed statistical-MLP neural networks, including an input layer, several hidden layers, and an output layer. The input layer takes the statistical features of path loss data into the neural network while the output layer produces the prediction for GPS spoofing attacks. In hidden layers, each layer contains a set of neurons that are connected with the next layer through the nonlinear activation function. Mathematically, the activation function is defined as:

$$y(t) = f(\sum_{x\in\mathscr{F}} \omega^{x}_{iu} \Delta L^{x}_{iu}(t) + T_x), \tag{4.2}$$

where $f(\cdot)$ is the nonlinear activation function, $\Delta L^{x}_{iu}(t)$ stands for the difference of path losses statistical features measured by the i^{th} BS at time t, and x belongs to $\mathscr{F}$ that consists of the mean (M), Variance (V), Skewness (S), Kurtosis (K), the minimum ($Q0$), the first quartiles ($Q1$), the sample median ($Q2$), the third quartiles ($Q3$), the maximum ($Q4$) and Wasserstein Distance (W) of actual path losses in the time interval t, $\mathscr{F} = \{M,V,S,K,Q0,Q1,Q2,Q3,Q4,W\}$. $y(t)$ denotes the output vector, $y(t) \in [0,1]$ and $y(t)$ closes to 1 indicating a higher likelihood of GPS spoofing attacks, specifically. The weight relationship among different statistical metrics is stored at ω^{x}_{iu} and the bias is T_x for the hypothesis testing. Here, the Mean, Variance, Minimum, Maximum, and Quartiles are fundamental statistical features that provide a broad perspective on the distribution of data. These measures can facilitate the detection of shifts in signal strength or variability, which might imply a spoofing attack. On the other hand, Skewness and Kurtosis are higher-order statistical features offering further insights into the shape of the data distribution. Variations in these features could indicate unnatural alterations of the signal, potentially indicative of a spoofing attack. Additionally, the Wasserstein Distance, a metric measuring the distance between two probability distributions, proves to be useful for comparing the distribution of signal measurements over time in the context of GNSS spoofing detection. Noteworthy changes in the Wasserstein distance could suggest a shift in the underlying data distribution, potentially signaling a spoofing attack. By leveraging statistical methods, the MLP can more effectively detect anomalous behavior from the inputs, thereby enabling it to make more accurate decisions.

Figure 4.3 shows another deep neural network named the CNN that takes grouped path loss data as input to predict the GPS spoofing attack. The grouped path loss data are processed through a series of layers, including the input layer, convolution layers, max pooling layer, flatten layer and fully-connected layers. Firstly, the input layer reshapes the collected path loss data into a path loss matrix and forwards it to the convolution layers. Within the convolutional layers, filters are employed to extract features from the raw path loss data and output the feature map for detecting spoofing patterns in the input data. Next, the feature map extracted from the convolution layers is passed through the max pooling layer, which reduces the spatial dimensions of the data and helps to capture the most important features. Finally, the output of the max pooling layer is flattened into a single vector and passed through fully-connected layers, or named dense layers, where the features are processed the same way as MLP which uses weights and biases to make predictions in the output layer. The different layers in the CNN work together to extract and refine the path loss features, which leads to accuracy improvement in the final GPS spoofing detection.

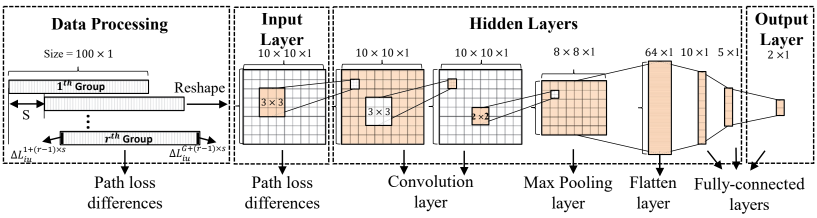


Figure 4.3. Convolution neural network for GPS spoofing detection

In Figure 4.3, $\mathscr{L}_{iu}^{1}$ stands for a group of path losses difference estimated from the i^{th} BS, and $\mathscr{L}_{iu}^{1}$ is:

$$\mathscr{L}_{iu}^{1} = \left\{\Delta L_{iu}^{1}, ..., \Delta L_{iu}^{g}, ..., \Delta L_{iu}^{G}\right\}, \tag{4.3}$$

where G denotes the group length, $G > 0$ and $\sqrt{G} \in \{1, 2, 3, ...\}$. Because the CNN can only solve the problem with spatially independent features [81], the detection results predicted by the CNN on each group regardless of the spoofed positions in the group. To overcome the CNN weakness, a step s is adopted for choosing a new group, which can help to improve the detection accuracy using grouped data, as seen in Figure 4.3. So that, the r^{th} $(r > 1)$ group contains G items and starts at $\Delta L_{iu}^{1+(r-1)\times s}$ and ends at $\Delta L_{iu}^{G+(r-1)\times s}$. In addition, these group data are reshaped as a $(\sqrt{G} \times \sqrt{G})$ matrix before feeding to the CNN.

Both MLP- and CNN-based GPS spoofing detection are built on deep neural networks that require data for optimizing the network parameters. The volume of the training data depends not only on the quality of the data but also on the complexity of the network. Compared with the CNN, the MLP has a simpler structure but needs additional statistical processing on the path losses data. On the contrary, the CNN can work on the raw path loss data directly but has a more complex network architecture. To mitigate the limitations of each network's structure, transfer learning, a technique that leverages knowledge from previously trained models, is introduced in the following subsection [82], and the performance comparison of the MLP and CNN is shown in the next section.

4.2.4 Transfer Learning Approach

Transfer learning-based GPS spoofing detection procedures are presented in Figure 4.4, where one BS can share a pre-trained model with the other BS over the edge servers. Firstly, a model M_i is trained on the edge server using the path loss differences data measured by BS i. Then, instead of transmitting a large amount of data over the network, BS i sends a pre-trained model with feature knowledge of path loss data to the next edge server. Finally, once it has received the model, BS $i+1$ can train its own model on the edge server using the pre-trained model M_i and its own path losses data. On one hand, transfer learning can help reduce the training time due to both the new model and the pre-trained model containing the same convolution layers for path loss feature extraction. On the other hand, transfer learning can reduce traffic load over the network because it allows training a new model with less data through the pre-trained model.

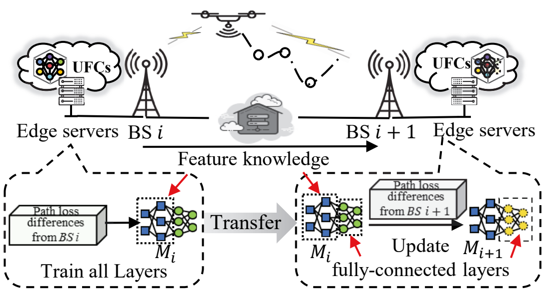


Figure 4.4. Transfer learning-based GPS spoofing detection

Even though transfer learning provides a representative and efficient model training approach, the trained models detect GPS spoofing independently inside each cell. In the handover area, these models can work together and further improve detection performance. By combining the detection results from multiple models, the overall accuracy of GPS spoofing detection can be increased, especially in a highly dynamic cellular-connected UAV environment. Noteworthy, transfer learning is only avail-

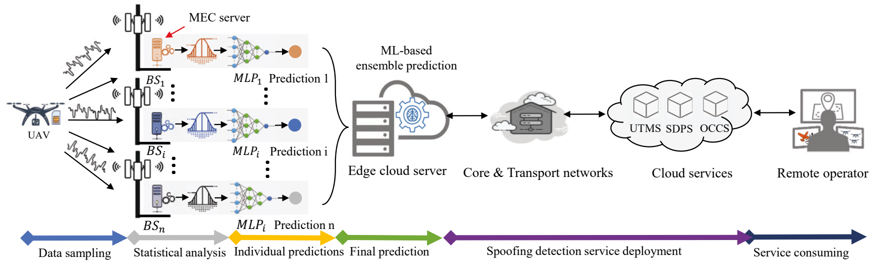


Figure 4.5. Deep ensemble learning-based GPS spoofing detection

able for CNN models that can share the same convolutional layers for feature extraction.

4.2.5 Deep Ensemble Learning Approach

Figure. 4.5 illustrates an MEC architecture, where deep ensemble learning is used to aggregate GPS spoofing detection results from different BSs. Deep ensemble learning includes four main processes, namely data sampling, statistical analysis, individual prediction, and final prediction. In addition, there are two extra processes in Figure. 4.5 for service deployment and consumption. The first three processes are the same as the statistical MLP provides the individual prediction and then forwards to the edge cloud server for integration. Finally, ensemble learning is executed by the edge cloud server to predict whether the GPS position is spoofed or not. The spoofing detection service is deployed on MEC servers by UTMS and consumed by the remote operator through OCCS, specifically.

There are two ensemble methods, namely threshold-based weighted aggregation and ML-based aggregation. The threshold-based weighted aggregation produces the final results by comparing a preset threshold with a weighted combination of individual predictions, while ML-based aggregation combines individual predictions through different machine learning techniques. However, it is a challenge to find the best threshold for the desired performance, which depends on various factors, such as the number of BSs, and false negative and false positive trade-offs. Compared with threshold-based weighted aggregation, ML-based aggregation can automatically learn the decision threshold from the data set, these machine learning methods including MLP, Decision Trees (DTs), Logistic Regression (LR), Support Vector Machine Kernel (SVMK), Support Vector Machine Gamma (SVMG), and Gaussian Naive Bayes (GNB) [83, 84, 85, 86, 87]. The performance results of threshold-based weighted aggregation and ML-based aggregation are summarized in the next section.

4.3 Performance Evaluation

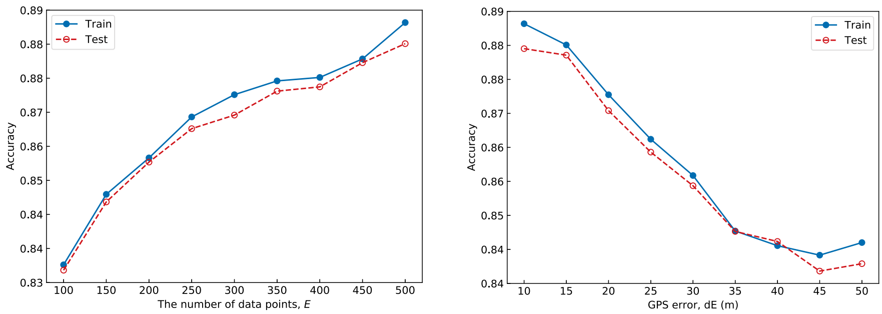
In this section, we evaluate the performances of the MLP, CNN, transfer learning, the threshold-based multi-MLP ensemble approach, and ML-based multi-MLP ensemble approach in terms of GPS spoofing detection. To this end, we develop a simulator using Python 3.7 and Tensorflow 2.7, where Python is used to develop a simulation platform and Tensorflow is applied to test the ML algorithms.

The simulation platform contains nine BSs evenly distributed in an area of 300×300 m^2, where each BS maintains a distance of 150 m from its neighboring BSs and all BSs are equipped with a 35 m antenna. The victim UAV, u, is above the center of the area with an altitude of 150 m. Initially, a total of 12 potential trajectories toward different directions are implemented to simulate the spoofing attacks. Specifically, a polling strategy is adopted to select two alternative paths from those twelve paths, one representing the planned path while the other standing for the spoofed path. Meanwhile, the simulation platform uses the path loss models defined in [80] with channel frequency 2.0 GHz to obtain the training data set from the BSs to u.

The performance of the MLP depends on hyperparameter settings, such as learning rate, the number of hidden layers, and the number of neurons per hidden layer [88]. To find the best configuration for the MLP model, the grid search is used on hyperparameter tuning, including the learning rate drawing from $\{0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001\}$, and the number of hidden layers taking from 1 to 6 layers and the neurons of each hidden layer varying from $\{8, 16, 32, 64, 96, 128\}$. In addition, the Rectified Linear Unit (ReLU) is used as an activation function for MLP. Early stopping with the patience of 15 is applied to prevent the MLP from overfitting on 2259 samples during at most 500 epochs. the MLP hyperparameter settings are summarized in Publication II.

The structure of the CNN model is shown in Figure 4.3, where a set of layers and a kernel are used to conduct GPS spoofing detection. The CNN input layer takes a tensor with size $\sqrt{G} \times \sqrt{G} \times 1$, where $\sqrt{G}$ is drawn from 5 and 30. Then, the convolution layers use a 3×3 kernel to extract features from the raw data. Following this, the pooling layer with a 2×2 kernel is employed to reduce the dimension of the neural network size. In the end, a flattened layer and 3 fully-connected layers work together for GPS spoofing detection. To prevent the CNN from overfitting, the dropout is set as 0.1 on each layer and along with early stopping with the patience of 20 is applied. The optimizer $RMSprop$ and the activation function $ReLU$ are used by all layers except the output layer that has a $softmax$ activation function. The CNN hyperparameter settings are summarized in Publication III.

The ensemble learning approach consists of the thresholds method and six kinds of machine learning methods. The threshold is chosen from 0



(a) The effect of E on accuracy under $dE = 15m$

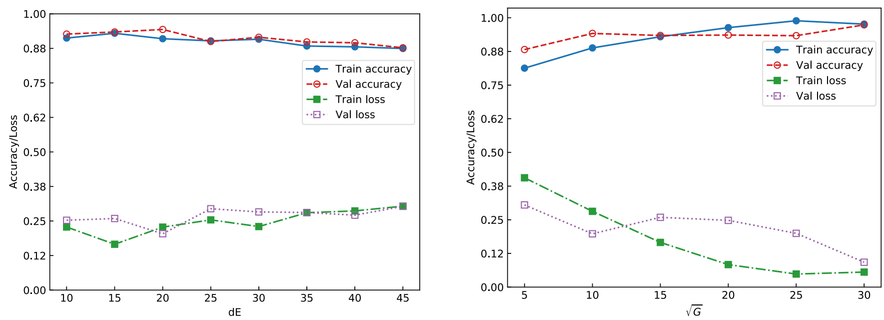
(b) The effect of dE on accuracy under $E = 150$

Figure 4.6. The accuracy performance of the statistical-MLP algorithm

to 1 with step 0.1. In addition, these thresholds are evaluated with the number of BSs ranging from 1 to 9. In addition, the False Positive (FP) and False Negative (FN) are the measurements for finding the best threshold. For the machine learning ensemble approaches, the MLP hidden layers contain n neurons that are the same as BSs' numbers, and the DT has the same depth ρ with the number of BSs n. To overcome overfitting in the logistic regression, the inverse regularization strength of LR is set as 0.01. To increase robustness and avoid overfitting, the linear kernel of the SVMK model is set a penalty coefficient $\sigma = 0.025$ while the radial basis function kernel of the SVMG uses parameters $gamma = 2$ and $\sigma = 1$, where $gamma$ decides the curative in the decision boundary and a bigger $gamma$ leads to a higher curvature value. The deep ensemble learning settings are summarized in Publication IV.

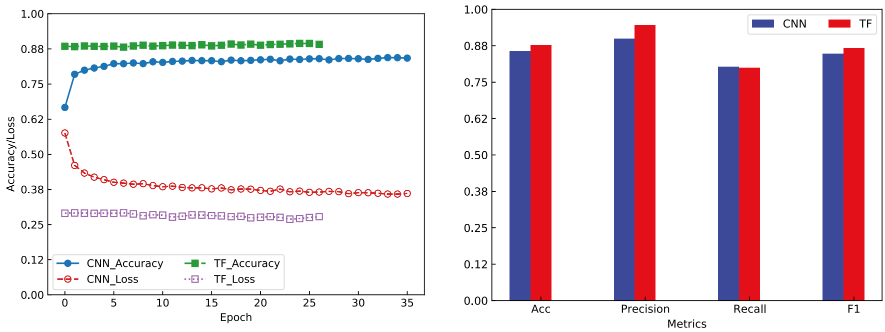
4.3.1 Statistical-MLP Approach

The performance of statistical-MLP-based GPS spoofing detection is shown in Figure 4.6, where we investigate both the number of data points E and the setting of GPS error dE impacts on MLP detection accuracy. Figure. 4.6a illustrates the MLP accuracy increases as E increases while the opposing trends are obtained from Figure. 4.6b that demonstrates the MLP accuracy decreases as dE increases. The former increasing phenomenon is a larger data set that brings higher statistical accuracy and precision, which then reflects on the MLP output performance. The latter decreasing phenomenon is a bigger GPS error in the higher dynamic environment, which can cover GPS spoofing attacks and make it easier for an attacker to counterfeit the GPS signals. Another observation from Figure. 4.6a and Figure. 4.6b indicates that the MLP shows a small difference but the same increasing/decreasing trend on both the train and test data set, which means the MLP model is well trained without overfitting. In addition, it is notable that the proposed statistical-MLP approach achieves more than



(a) GPS error(dE), $\sqrt{G} = 15$ **(b)** Group size($\sqrt{G} \times \sqrt{G}$), $dE = 15$

Figure 4.7. The accuracy and loss of the CNN with different settings



(a) Training history **(b)** Performance metrics

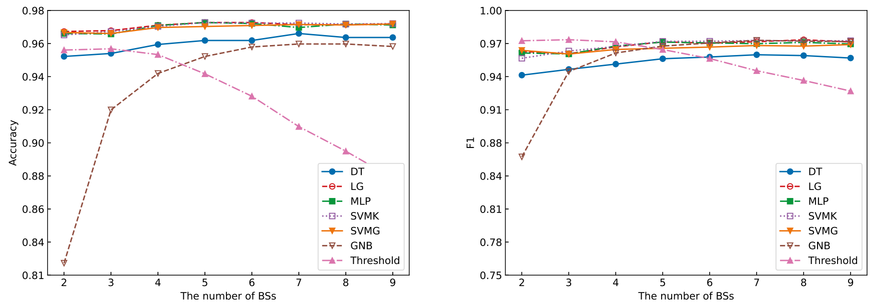
Figure 4.8. Training history and performance recording

0.83 detection accuracy on 1×10^6 data points with a split of 0.3 for testing while 0.7 for training.

4.3.2 CNN and Transfer Learning Approach

Figure 4.7 and 4.8 shows the GPS spoofing detection performance of the proposed CNN and transfer learning models in terms of accuracy and sparse categorical cross-entropy. The performance metrics in Figure 4.7 and Figure 4.8 are based on 1.5 million data points from 16 trajectories, with one selected as planned and another from the remaining 15 chosen as spoofed in a rolling manner; the tolerated GPS margin error was adjusted in steps of 5 meters from 10 to 45 meters, and data was reshaped into groups ranging from 5×5 to 30×30 with step 10 for network input. The validation split is set as 0.3 for the testing data set.

In Figure 4.7, the CNN model is evaluated by varying the GPS margin error d_E from 10 meters to 45 meters and the group size $\sqrt{G}$ from 5 to 30. It is noted from Figure 4.7a that CNN detection accuracy is insignificantly decreasing with increasing dE because a bigger GPS error makes it difficult to distinguish the trajectory deviation caused by a highly dynamic



(a) The effect of the number of BSs on accuracy

(b) The effect of the number of BSs on F1

Figure 4.9. The performances of ensemble methods under $dE = 15m$ and $E = 150$

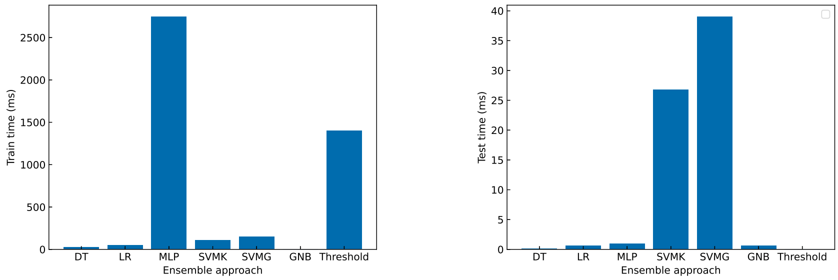
environment or GPS spoofing attacks. On the contrary, it is observed from Figure. 4.7b that the bigger group size achieves a higher accuracy, which is due to the more elements in the group providing the CNN model with more path losses data.

Figure 4.8 illustrates the performance comparison between a normally trained CNN and a transfer learning-trained CNN. Figure. 4.8a indicates that transfer learning can leverage knowledge in a pre-trained model to improve model accuracy as well as reduce model training time. Indeed, the results depicted in Figure. 4.8b interprets that the model trained with transfer learning achieves a performance gain in terms of accuracy, precision, and F1 score. Thus, transfer learning is a promising solution for the edge to detect GPS spoofing.

4.3.3 Deep Ensemble Learning Approach

To evaluate the effectiveness of threshold-based and ML-based ensemble methods in detecting GPS spoofing attacks, we measure *Accuracy* and *F1-score* by turning the number of BSs and using 10^6 path losses data, where 70% data is for model training and 30% data is for model testing, as shown in Figure 4.9.

Figure 4.9a shows that ensemble methods can significantly improve spoofing detection performance in terms of *Accuracy* and *F1-score*. For instance, the spoofing detection accuracy is nearly 0.97 with two BSs using ML-ensemble methods, while it is less than 0.90 using the statistical-MLP method. The reason is that ensemble methods take more data from different BSs at the same time, which enriches the evidence for making a decision. The decline in accuracy observed in Figure 4.9a with increasing base stations (BS) may be due to system complexity and the limitations of the threshold algorithm. More BSs lead to increased signal interference and noise, complicating GNSS spoofing detection. The same conclusion is



(a) Train time **(b)** Test time

Figure 4.10. Training and testing time consumption of ensemble algorithms

also found in Figure 4.9b. Furthermore, when turning up the number of BSs, ML-ensemble methods show a better performance than the threshold method on both *Accuracy* and *F1-score*. The reason for this phenomenon can be attributed to the inherent adaptability and flexibility of machine learning methods, which enable them to effectively handle various situations, including those that are either represented in the training data set or significantly deviate from the non-spoofed data set. Moreover, the GNB-ensemble method exhibits the worst performance compared with the other ML-ensemble methods, because the GNB makes an unnecessary assumption that the path losses data need to follow the normal distribution and therefore leads to an incorrect decision.

We further compare the time spent by ML-based ensemble methods and threshold-based ensemble methods on the same machine in Figure 4.10. The training time refers to the duration required either for training the machine learning model or for determining the optimal threshold value using the same data set. The optimal value is characterized by minimizing missed detections while also taking false alarms into account. Notably, missed detections are considerably more harmful to the system than false alarms, as they can lead to violations of no-fly zone regulations and increased risks of collisions. A comprehensive summary of this can be found in Figure 4.10a. Figure 4.10a indicates that it takes more time to train the MLP model than the others due to the MLP being a deep neural network and containing more neurons than other machine learning models. Noteworthy, an optimal threshold can be utilized as a reference for a different data set; however, this optimal threshold may result in higher detection errors in the new data set. Figure 4.10a shows SVMK and SVMG require more time on the test data set because these two models are not incremental and therefore must analyze all the test data at once. Regarding the threshold-based ensemble method, it consumes around 1400ms on finding the best threshold, which is higher than most ML-based ensemble methods. For the testing time, the threshold-based ensemble methods detect the spoofing by comparing the test values with the thresholds directly, which has the lowest complexity of all ensemble methods.

Based on Figure 4.9 and 4.10, we find that the ML-based ensemble methods can achieve a trade-off between accuracy and efficiency. Meanwhile, a flexible combination of deep learning and other machine learning methods could improve detection accuracy and reduce detection latency.

4.4 Summary and Discussions

This chapter explored deep neural networks-based GPS spoofing detection approaches for cellular-connected UAVs. To detect GPS spoofing when the UAV is located in the area covered by a single BS, an MLP neural network is designed to analyze the statistical path loss values and is able to detect the UAV trajectory deviation caused by spoofing attacks. In addition to the MLP, another deep neural network, named the CNN, is designed to analyze the raw path loss data directly to reduce the detection latency and increase the detection accuracy. Indeed, for GPS spoofing detection, the CNN is recommended over the MLP. This is because the CNN can effectively process the temporal structure of path loss data and uncover key spoofing patterns that the MLP might not capture as efficiently. Thus, with sufficient computational resources, the CNN emerges as a more effective choice. Specifically, transfer learning is introduced to transfer the CNN knowledge between edge servers from one BS to another, which can reduce the CNN training time and increase the CNN detection accuracy. To detect GPS spoofing when the UAV is located in an area covered by multi-BSs, deep ensemble learning is proposed to ensemble the individual edge server MLP results. Therefore, the use of a combination of different neural networks can consecutively monitor the UAV positions and detect GPS spoofing for cellular-connected UAVs.

The proposed deep learning methods allow us to detect GPS spoofing attacks using a single BS in an energy-efficient way for cellular-connected UAVs. The developed models are trained and tested with artificial simulated path losses data that are driven from a statistical model defined in [80] by 3GPP. Generally, the path loss estimation contains significant errors in the high dynamic UAV communication channel. For a better evaluation, the models should be trained and evaluated with real data in order to immune environmental impacts. In addition, these models are deployed in different edge servers for GPS spoofing detection, one model per UAV specifically, which may cause congestion in the detection system due to the limited computation resources in the edge server. Results have shown that the proposed MLP model can achieve detection accuracy of more than 88% using 500 path loss data under settings with one BS and 15 meters GPS error, while the CNN model has a comparable performance using only 225 path loss data. Although these models can protect cellular-connected UAVs from GPS spoofing attacks, they are no longer efficient in a situation where a flock of UAVs is deployed at the same time for cooperative tasks. To this end, a new type of Graph Neural Network (GNN) is introduced in the next chapter to deliver an efficient GPS spoofing detection for cellular-connected UAV swarms. It is important to emphasize that the time required for detection and the time interval between false alarms and the distance deviated from the reference path are crucial metrics in evaluating spoofing detection performance. In the context of transfer learning methods, the time taken to deploy the machine learning model on the edge server could result in substantial detection latency. Moreover, the deep ensemble learning method requires collaboration among multiple base stations, making it susceptible to network latency. Such detection latency has the potential to trigger an incident in the event of a spoofing attack. In the future, we will include detection time as a key performance indicator for the system.

5. GNN based GPS Spoofing Detection for Cellular-Connected UAV Swarms

This chapter proposes a novel GPS spoofing detection approach for a cellular-connected UAV swarm using Graphic Neural Network (GNN) layers and recurrent network layers, where the GNN computes the spatial similarity between the swarm GPS topology and communications topology while the recurrent network layers acquire the temporal similarities of the swarm topology changes caused by GPS spoofing attacks. Precisely, the vertex of the swarm GPS topology is the UAVs' GPS coordinates, while the edge of the swarm communications topology is the wireless connection links. The contribution of this chapter is based on Publication V and VI.

5.1 Problem Description

Unmanned Aerial Vehicles (UAVs) are emerging as an integral part of the 5G-and-beyond system due to their deployment and movement flexibility. Compared with the traditional single UAV, a flock of UAVs established as an aerial swarm can supplement temporary network connections and implement diverse distributed applications economically and efficiently [89]. However, the GPS spoofing attacks are able to manipulate aerial swarm locations and distort aerial swarm topology, which threats the security of swarm communication and control [40]. In fact, the cellular-connected UAV swarm is formatted as a Flying Ad-hoc Network (FANET) system that uses an Ad-hoc On-demand Distance Vector (AODV) routing protocol for communication and control. Because the AODV protocol has less protection and authentication on the node, it makes the swarm FANET vulnerable to a spoofing attack [90]. Although the proposed deep learning and transfer learning approaches demonstrate effectiveness in detecting GPS spoofing, they are designed for one single cellular-connected UAV and do not consider the case of a cellular-connected UAV swarm. Nevertheless, it is not efficient to use the aforementioned deep learning methods to detect spoofing for a UAV swarm by running multi-models on edge servers, one per UAV. There are two main issues with GPS spoofing detection for cellular-connected UAV swarms.

1. First, rather than analyze every single UAV signals, an efficient way is one that leverages the swarm GPS topology information and the swarm communication topology to detect spoofing attacks for cellular-connected UAV swarms. The Graph Edit Distance (GED) is used to compare two graph-pairwise similarities. However, the conventional GED is an NP-Hard problem and cannot compare a graphs-pair with more than 16 nodes in a reasonable time [91]. Thus, it requires an efficient GED computation method.

2. Second, the GED can describe the spatial similarity of graph-pairwise. However, the UAV swarm topology is dynamic during spoofing attacks. Therefore, the temporal similarity changes are important evidence to detect abnormal behaviors of the swarm caused by spoofing attacks.

The proposed GNN-based GPS spoofing detection solutions are described in the following sections.

5.2 Modeling and Proposed Solutions

In this section, we first introduce the cellular-connected UAV swarm system. Then, the problem of GPS spoofing attacks on cellular-connected UAV swarms is reviewed and formulated mathematically. After that, the GNN and RNN are presented to detect GPS spoofing for cellular-connected UAV swarms.

5.2.1 Cellular-Connected UAV Swarm

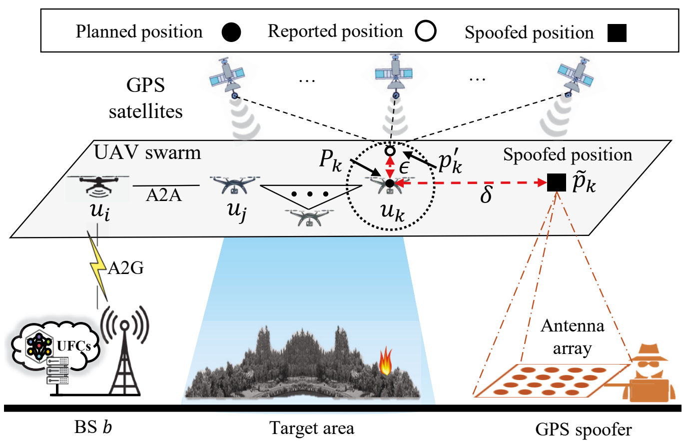


Figure 5.1. Cellular-connected UAV Swarm

Figure 5.1 illustrates a cellular-connected UAV swarm system that consists of a BS b, a UAV swarm including N UAVs, and a GPS spoofer with an antenna array. The BS is connected with the edge UFCs in charge of the swarm control and monitoring. The cellular-connected UAV swarm contains one swarm leader u_i and the other swarm members, for example, u_j and u_k, $0 < i,j,k \leq N$ and $i \neq j \neq k$, where the swarm leader is directly connected with the BS through an Air-to-Ground (A2G) link while the swarm members are connected with the leader over Air-to-Air (A2A) links. Both A2G and A2A are defined in [80] by the 3GPP. The GPS spoofer uses its antenna array to send fake GPS signals and lead the UAV to the wrong location, which can manipulate UAV positions. For instances, u_k reports its position at $p_k^{'}$ that is close to its planned position p_k but its real position is at at $\tilde{p}_k$ that deviates from p_k with δ because of GPS spoofing, $\epsilon \leq dE < \delta$, where ϵ denotes the GPS error between p_k and $p_k^{'}$ while dE represents the max GPS error.

5.2.2 Problem Formulation

Let $\mathscr{G}$ denote an undirected graph for the swarm formation, $\mathscr{G} = \{V, E, W\}$, where $V = \{v_0, ..., v_i, ..., v_N\}$ is the graph node set and v_0 stands for the BS node while v_i represents the i^{th} UAV node, E is edge set and $E \subseteq \{(v_i, v_j) : v_i, v_j \in V, 0 \leq i \neq j \leq N\}$ particularly, W is the weighted adjacency matrix that consists of the interaction strength, $W = [w_{ij}] \in \mathbb{R}^{(N+1)\times(N+1)}(w_{ij} \geq 0)$. In practice, the swarm formation can be obtained from GPS positions or from wireless connections, and let $\mathscr{G}_g$ indicate the GPS positions formation while $\mathscr{G}_l$ express the wireless connections formation. The interaction strength in $\mathscr{G}_g$ is related to the distance between two nodes and correspondingly it in $\mathscr{G}_l$ stands for the path loss value from node i to node j. Theoretically, $\mathscr{G}_l$ is similar to $\mathscr{G}_g$ because communication path losses between the transmitter and receiver depend on their distances. However, the GPS spoofer fools the UAV GPS receiver compute fake positions, which brings an incorrect distance computation as well as disrupts the corresponding relationship between $\mathscr{G}_g$ and $\mathscr{G}_l$. Therefore, the difference between $\mathscr{G}_l$ and $\mathscr{G}_g$ can indicate GPS spoofing in the cellular-connected UAV system. Traditionally, the GED between $\mathscr{G}_g$ and $\mathscr{G}_l$ quantifies the minimum number of edit operations, such as vertex or edge insertion and deletion, that are necessary to transform $\mathscr{G}_g$ to $\mathscr{G}_l$ [92]. Let $\mathfrak{S}_{gl}$ denote the difference between $\mathscr{G}_l$ and $\mathscr{G}_g$, and the spoofing detection problem can be formulated as:

$$\begin{cases} H_0: & \mathfrak{S}_{gl} > \tau, \\ H_1: & \mathfrak{S}_{gl} \leq \tau, \end{cases} \tag{5.1}$$

where τ is the threshold for hypothesis testing, and H_0 presents GPS spoofing while H_1 standing for the opposite. The GPS is considered as spoofed when $\mathfrak{S}_{gl}$ is above the threshold τ. Otherwise, H_1 is accepted if $\mathscr{G}_l$ is similar with $\mathscr{G}_g$. However, it is difficult to compute the graph distance $\mathfrak{S}_{gl}$ using the conventional GED that is defined as the number of edit operations to transform one graph into another, including graph edge deletion, edge insertion, and vertex relabeling [93]. Although the GED is the most popular metric for evaluating pairwise graph similarity, it is an NP-Hard problem and cannot compare a graphs-pair with more than 16 nodes in a reasonable time [91]. Thus, those conventional methods are invisible to compute the similarity between the swarm GPS topology and communication topology because of too much time consume, which may lead to intolerable delay and hinder the spoofing detection.

5.2.3 GNN-based GPS Spoofing Detection Approach

To address the aforementioned challenges, we employ the GNN and RNN models in the edge server to devise an effective GPS spoofing detection approach for the cellular-connected aerial swarm. The GNN is used for computing the similarity between two graphs with acceptable time consumption while the RNN is adopted for spoofing detection by analyzing the dynamic changes in graphical similarity.

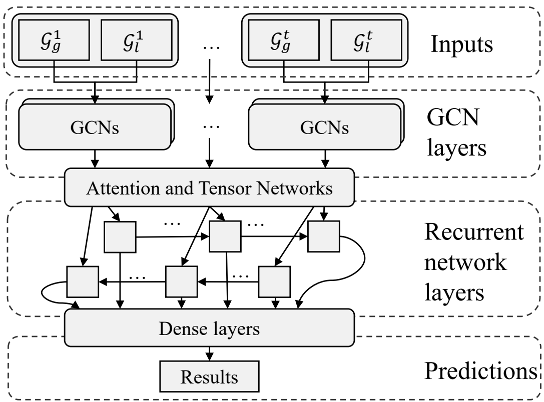
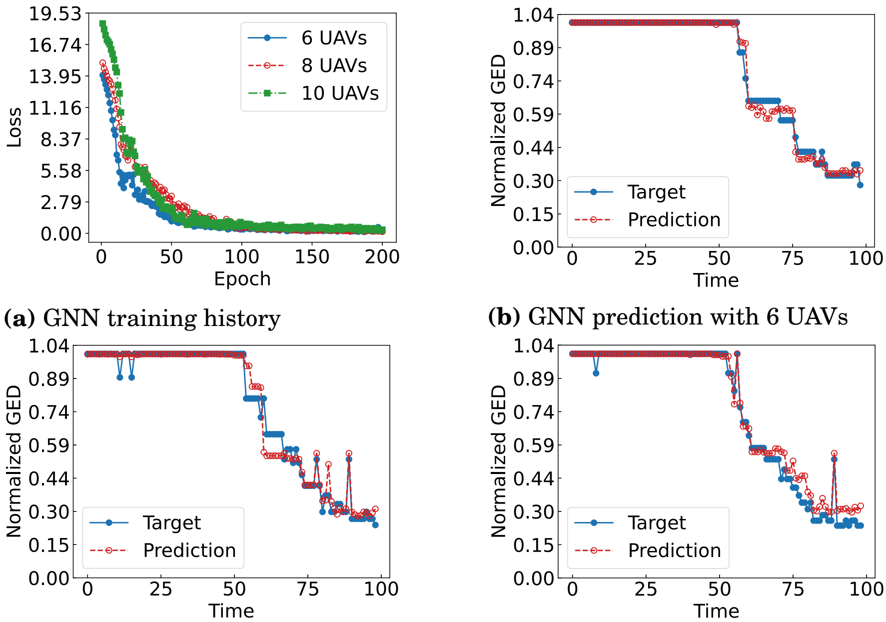


Figure 5.2. GNN architecture

As illustrated in Figure 5.2, the GNN uses the swarm GPS topology $\mathscr{G}_g$ and the swarm communication topology $\mathscr{G}_l$ as inputs and produces the similarity score $Similarity(\mathscr{G}_g, \mathscr{G}_l)$ between $\mathscr{G}_g$ and $\mathscr{G}_l$. In the GNN, the Graph Convolutional Networks (GCNs) layers are used to embed the swarm GPS topology and one of its nodes into the Euclidean vector space [94]. Precisely, all $N+1$ nodes in a graph $\mathscr{G}$ have $N+1$ node embeddings. The attention and tensor network layers are in charge of graph-level embeddings that aggregate all of the nodes' embeddings into one graph embedding with the help of a novel attention mechanism, where the attention mechanism emphasizes the node importance and node similarity [95]. However, the



(a) GNN training history

(b) GNN prediction with 6 UAVs

(c) GNN prediction with 8 UAVs

(d) GNN prediction with 10 UAVs

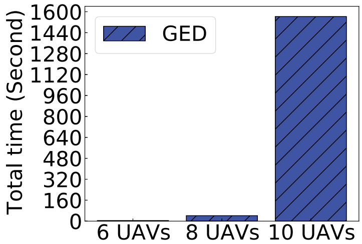
Figure 5.3. GNN training history and prediction

attention emphasizes the graph embeddings but loses the small substructure features, which debilitates the authenticity of the final similarity [96]. To solve this problem, the Neural Tensor Network (NTN) layers are employed for extracting fine-grained node-level information. Although the GNN-based approach can improve the efficiency of the spatial similarity comparison between two graphs in a given time, it cannot capture the swarm topological changes caused by GPS spoofing attacks over a long time [97]. In this vein, it is necessary to add the recurrent neural network (RNN) layers at the end of the GNN in order to acquire the GPS spoofing temporal features and improve the detection performance.

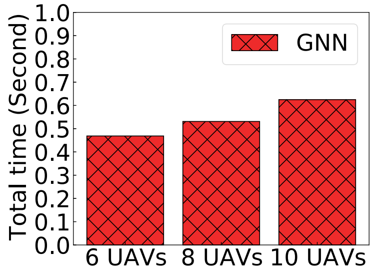
5.3 Performance Evaluation

We develop a swarm simulator using Python 3.6 and PyTorch 1.1.0 to evaluate the performances of the proposed spoofing detection algorithms. In this simulator, Python is used to set up the simulation platform, and PyTorch is employed to run the GNN model. The simulation platform contains one base station and an aerial UAV swarm in a 3D $1400 \times 1400 \times 100$ m^3 space. In addition, the channel frequency is set to 5.0 Ghz for A2A communication and 2.4 Ghz for A2G communication. Detailed parameters are shown in Publication VI.

Figure 5.3 depicts the GNN training history and prediction results for

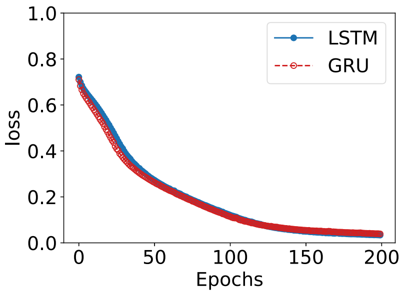


(a) GED computation time

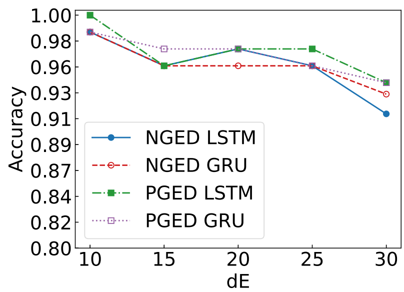


(b) GNN computation time

Figure 5.4. Computation times for 100 graphs pairwise



(a) Training history



(b) Detection performance

Figure 5.5. Training history and detection performance for 10 UAVs

6, 8, and 10 UAVs in a UAV swarm. Figure 5.3a shows the GNN training history, where the GNN can fit the swarm topology and compute the similarity with a small loss value. Referring to Figure 5.3a, the loss value, which demonstrates the accumulation of errors in the GNN model during the training, exhibits a higher value at 50 epochs for 8 UAVs compared to 10 UAVs. This outcome can be attributed to the fact that neural networks typically commence with random weight initialization, which could influence the learning process. Figure 5.3b, 5.3c and 5.3d proves the trained GNN performs well on the test data set and keeps a good prediction on graph similarity, where the y-labels is the normalized GED within a standardized range of [0,1] [96].

Figure 5.4 illustrates the time spent by GED methods and GNN methods for computing the 100 graph-pairwise similarities. Compared with traditional GED methods that compute the graph similarity by counting edition operations, the GNN is 1000 times faster than the GED in terms of graph-pairwise similarity computation because it computes the similarity directly, thanks to different kinds of neural network layers that embed the graph into Euclidean vector space. Remarkably, the GNN uses less than 10 ms for the graph pairwise with 10 nodes, which is almost 1600 seconds for the traditional GED method.

Figure 5.5 demonstrates the final GNN-based GPS spoofing detection for

10 UAVs, which tests the performance of the Normalized GED (NGED) and the Predicted GED (PGED) under different GPS errors. It is obtained from Figure 5.5a that the GRU is convergent faster than the LSTM network because the GRU model has a more simple architecture compared with the LSTM. In Figure 5.5b, GNN accuracy is decreasing with increasing GPS error, which is due to the fact that it is easier to counterfeit the GPS signals in a highly dynamic environment. In addition, the accuracy of the NGED is close to that of the PGED, which means that the GNN has the safe efficiency as the GED for GPS spoofing detection. Moreover, although the LSTM shows comparable performance, the GRU spends less time on the model training which can reduce detection latency in a dynamic environment.

5.4 Summary and Discussions

This chapter introduced the GNN-based GPS spoofing detection approach for the cellular-connected UAV swarm. the GNN first uses its own layers and the layers of other neural networks to compute the spatial similarity between the swarm GPS topology and communications topology. Then, the GNN uses recurrent network layers to compute the temporal similarities of the swarm topology changes caused by GPS spoofing attacks. In the results, the combination of GCN layers with GRU layers shows the smallest training time and is the most efficient detection neural network model for cellular-connected UAV swarms.

The use of the GNN demonstrates its effectiveness in detecting GPS spoofing for the cellular-connected UAV swarm. Compared with the conventional GED computation method, the developed GNN models compute the graphs' similarity in a different way by embedding the graph into a vector. The GNN model is trained and evaluated with a simulated data set that only takes into account the perfect swarm communication topology without considering communication errors. For a better evaluation, the inputs of the GNN model should be added extra links or removed parts of nodes in order to incorporate the swarm communication errors. The simulation results show that setting the UAV swarm with 10 UAVs GNN can compute the swarm topology similarity within 1 second while it takes almost 1600 seconds for the state-of-the-art GED method. However, the GNN can not recover the swarm position after GPS spoofing detection. In the next chapter, we will elaborate more on ML-based GPS spoofing mitigation methods for cellular-connected UAVs.

6. 3D Radio Map-based GPS Spoofing Detection and Mitigation for Cellular-Connected UAVs

This chapter leverages a 3D radio map and machine learning methods to detect and mitigate GPS spoofing attacks for cellular-connected UAVs. Precisely, the edge UAV flight controller uses ray tracing tools and deterministic channel models to construct a 3D radio map. Then the machine learning methods are employed to detect GPS spoofing using that the 3D radio map. Once spoofing is detected, the particle filter is applied to relocate the UAV and mitigate GPS deviation. The contribution of this chapter is based on Publication VII.

6.1 Problem Description

The aforementioned machine learning approaches demonstrate effectiveness in detecting GPS spoofing attacks for cellular-connected UAVs. It is essential to recover the real GPS position after spoofing detection. Although the recent work in [59] enables the UAV to recover the real GPS signals from the spoofed ones, its performance can be further improved by using a radio map and machine learning approaches, where the radio map can provide the nutrition to a machine learning model that detects and mitigates GPS spoofing attacks. There are two main issues addressed in this chapter.

1. Radio map is constructed on ray tracing and the deterministic radio propagation model. However, it takes too much time to perform ray tracing in 3D space for cellular-connected UAVs. Thereby, an efficient 3D radio map construction method is required to detect GPS spoofing for cellular-connected UAVs in real time.

2. Detection is paramount to protect cellular-connected UAVs from GPS spoofing attacks, while mitigation is indispensable to recover and relocate UAVs after spoofing detection. Therefore, the detector should be able to mitigate the GPS spoofing attack using a 3D radio map.

State-of-the-art radio map construction methods include data-driven methods and model-driven methods. The former methods build a radio map that contains a large number of positions and electromagnetic data in a 3D space [98], while the latter methods build a radio map as a function of geographic locations [99, 100]. Data-driven methods require electromagnetic data to be evenly distributed, while model-driven methods need extra channel information [101]. However, it is difficult to build a radio map in an urban area where buildings are irregular and radio propagates complicatedly. Therefore, such radio map construction methods are not optimal solutions for GPS spoofing detection in an urban area.

6.2 Modeling and Proposed Solutions

In this section, we first introduce the cellular-connected UAV system in an urban area. Then, we provide a radio map model for GPS spoofing detection and mitigation. Finally, we demonstrate radio map-based GPS spoofing detection and mitigation methods.

6.2.1 Cellular-Connected UAV Network System in Urban Area

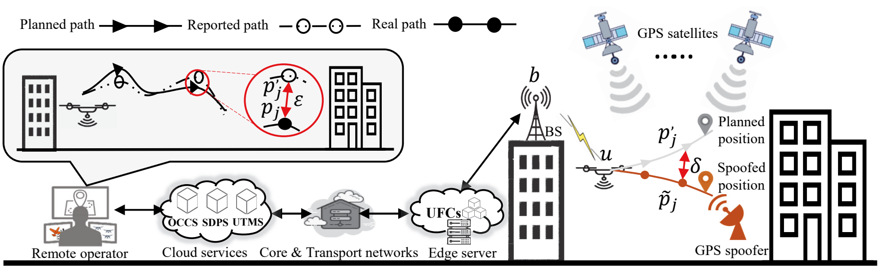


Figure 6.1. Cellular-connected UAV network system in urban area

Figure 6.1 illustrates the cellular-connected UAV network system in an urban area, where the remote operator uses the same network services as Publication III to control and monitor cellular-connected UAVs. In this urban area, the UAV u is located at p_j and is going to position $p_j^{'}$. However, the UAV arrives at $\tilde{p}_j$, having deviated from p_j with δ because of the GPS spoofing attack. Although the BS can detect trajectory deviation with the proposed deep learning methods, it cannot recover the real position with only a single BS. To mitigate GPS spoofing attacks, the radio map is introduced to relocate the UAV and recover the real position.

6.2.2 Radio Map Model

Let $\mathcal{M}_\tau$ denote a radio map that consists of K positions randomly distributed in a region $\mathcal{S}_\tau$, where $\mathcal{S}_\tau$ contains the UAV trajectory from p_j to

p_{j+1} specifically. Accordingly, a corresponding RSS value $\mathfrak{r}_k$ is appropriated to p_k, $p_k \in \mathscr{S}_\tau$ and $0 < k \leq K$. So, $\mathscr{M}_\tau$ is initiated as:

$$\mathscr{M}_\tau \leftarrow \{p_k, \mathfrak{r}_k\}, \forall k, k \in [1, K]. \tag{6.1}$$

$\mathscr{M}_\tau$ can provide the RSS for any position in $\mathscr{S}_\tau$. For instance, given a new location $p_i, p_i \in \mathscr{S}_\tau$, $\mathscr{M}_\tau$ can use the interpolation method to predict $\mathfrak{r}_i$ value that is close to the RSS values at nearby positions. Compared with the theoretical path losses model in the proposed deep learning approaches, the radio map can provide more accurate path loss values because it uses the deterministic channel model that considers the shape information of the buildings in an urban propagation environment. However, the deterministic channel model requires a large amount of ray tracing to find the direct path, roof edge diffraction paths, and wall reflection paths, which introduces excessive computational load to the edge server. Considering the limited resources of the edge server, the mission-based fine-grained 3D radio map is developed and detailed in Publication VII. The use of a radio map for GPS spoofing detection and mitigation is summarized in the following subsections.

6.2.3 Problem Formulation

The GPS spoofing detection problem is formulated the same as hypothesis testing defined in (4.1).

$$\begin{cases} H_0: & \Delta L_{bu} > T, \\ H_1: & \Delta L_{bu} \leq T, \end{cases} \tag{6.2}$$

where ΔL_{bu} is the absolute difference between theoretical path loss value obtained from the radio map and real path loss measured from the signal, $\Delta L_{bu} = \left|\tilde{L}_{bu} - L_{bu}\right|$. H_0 means GPS spoofing when the difference ΔL_{bu} is bigger than a preset threshold. Otherwise, H_0 is accepted as standing for GPS error. Compared with (4.1), the radio map can provide a more accurate theoretical path loss value to the reported position because the radio map is built on ray tracing and a deterministic channel model that considers the building shape in the urban area [102].

The GPS spoofing mitigation problem is formulated as an optimization problem.

$$\min_{\mathfrak{s}_1,\ldots,\mathfrak{s}_N} \quad \{\mathfrak{s}_n; n = 1, \ldots, N\}, \tag{6.3}$$

$$\text{Subject to:} \quad \mathfrak{s}_n = \Phi(\mathfrak{r}_{1:M}, \tilde{\mathfrak{r}}^n_{1:M}), \tag{6.3a}$$

$$0 \leq n \leqslant N, \tag{6.3b}$$

$$0 \leq \mathfrak{s}_n \leqslant 1; \tag{6.3c}$$

where $\mathfrak{r}_{1:M}$ are the real-time RSS values collected by the UAV from p_j to p_{j+1}, $\mathfrak{r}_{1:M} = \{\mathfrak{r}_i; i = 1, \ldots, M\}$ correspondingly. $\tilde{\mathfrak{r}}^n_{1:M}$ stands for the theoretical RSS provided by $\mathscr{M}_\tau$ for the path from p_j to $\tilde{p}_n$, where $\tilde{p}_n$ is one of the estimated positions where the UAV may be. $\tilde{p}_n$ belongs to $\mathbb{P}_N$, $\mathbb{P}_N = \left\{\tilde{p}_n; n = 1, \ldots, N\right\}$ respectively. $\mathfrak{s}_n$ is the similarity between $\mathfrak{r}_{1:M}$ and $\tilde{\mathfrak{r}}^n_{1:M}$. The target function aims to find the position that has similar RSS values on the estimated trajectory, which can be indicated by $\mathfrak{s}_n$, the Wasserstein distance between $\mathfrak{r}_{1:M}$ and $\tilde{\mathfrak{r}}^n_{1:M}$ [103]. It is worth noting that the choice of number N impacts the final location accuracy and the mitigation performance because a bigger N comes with higher accuracy but also a higher computation.

6.2.4 Radio Map-based GPS Spoofing Detection

The radio map-based GPS spoofing detection method is similar to the proposed deep learning methods, obtaining real and theoretical RSS values, computing RSS differences, data processing, and neural network analysis, as seen in Figure 6.2. All of these procedures run on the edge server.

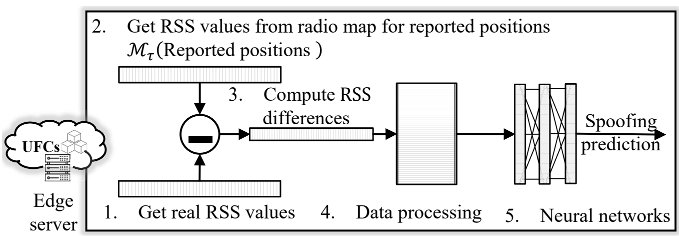
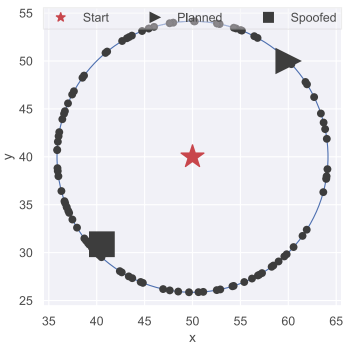


Figure 6.2. Radio map-based GPS spoofing detection

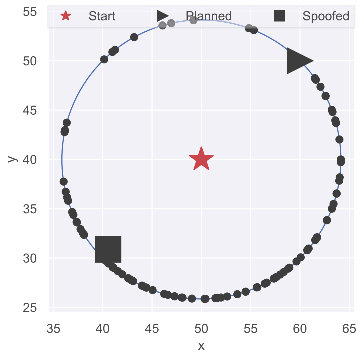
Firstly, the edge server obtains real RSS values from UAV telemetry data or from the BS. In an assumed manner, GPS spoofing attacks are not able to influence mobile cellular network communications. Secondly, the edge server uses a radio map to predict a theoretical RSS according to the UAV-reported GPS position. Thirdly, to solve the spoofing detection problem defined in (6.2), the edge server computes the difference between the real RSS value and the theoretical RSS value. A big RSS difference indicates the real RSS value breaking the law of signal propagation and thereby the reported GPS is spoofed. Fourthly, data processing formats the data with regard to the neural network requirement. Fifthly, neural networks use processed data to detect GPS spoofing.

6.2.5 Radio Map-based GPS Spoofing Mitigation

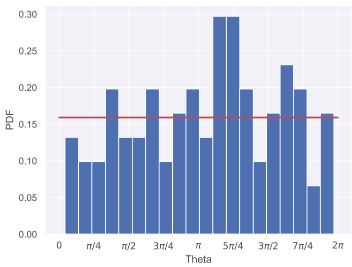
Radio map-based GPS spoofing mitigation relocates the UAV through a particle filter, which is a nonlinear estimator by using a set of particles or samples to represent the probability distribution of UAV's real posi-



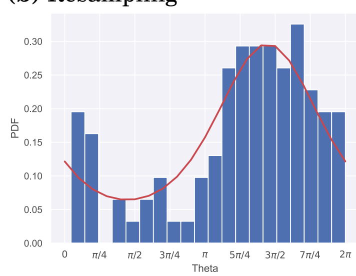
(a) Random sampling



(b) Resampling



(c) Random sampling PDF



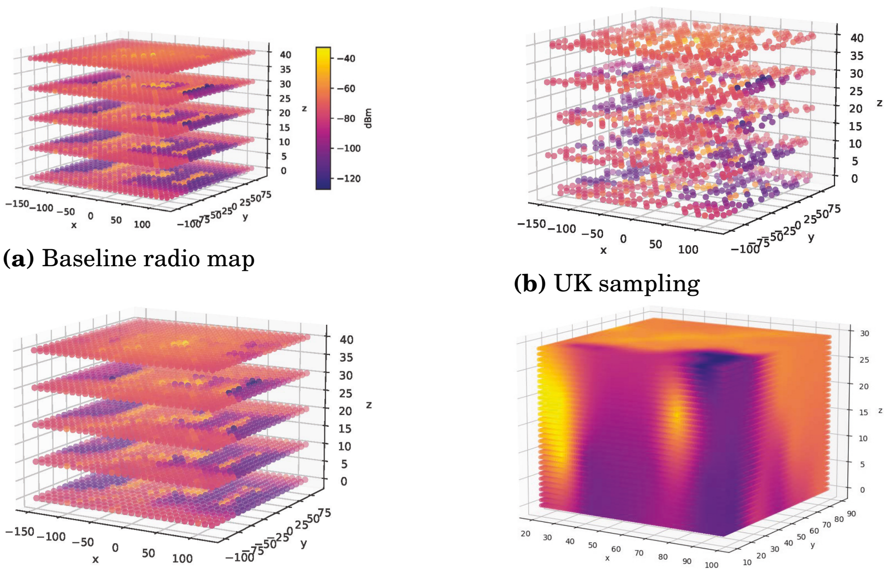
(d) Resampling PDF

Figure 6.3. Particle filter-based GPS spoofing mitigation

tions within the radio map. Compared with the EKF, another nonlinear estimator for localization, the Particle Filter is more adaptable and less assumptive about the system, while maintaining manageable computational demands on edge servers. With its ease of implementation, the particle filter efficiently determines the UAV's position through a two-step process comprised of sampling and resampling [104]. The sampling steps randomly choose N particles on the circle that is centered at the initial position p_j with radius d_{ij}, where d_{ij} is the distance recorded by IMU sensors from the initial position to the spoofing position. The reported position is close to the planned position in order to cheat the UAV GPS receiver, while the spoofed position is near to the real position, as shown in Figure 6.3. In addition, these N particles obey the Mises distribution that can random an angle φ from the circle with a Probability Density Function (PDF):

$$f(x \mid \mu_\varphi, \kappa_\varphi) = \frac{exp(\kappa_\varphi \cos(\varphi - \mu_\varphi))}{2\pi I_0(\kappa_\varphi)}, \tag{6.4}$$

where μ_φ and κ_φ denote the mean of the direction angle and the concentration of the direction angle, and $I_0(\kappa_\varphi)$ is the modified Bessel function for choosing angle φ [105]. In addition, $\mu_\varphi = 0, \kappa_\varphi = 0$ means that the random sampling follows a uniform distribution, and all of N particles are evenly distributed on the circle. To find the real position, a set of particles is



(a) Baseline radio map

(b) UK sampling

(c) UK interpolation

(d) UK fine-grained radio map

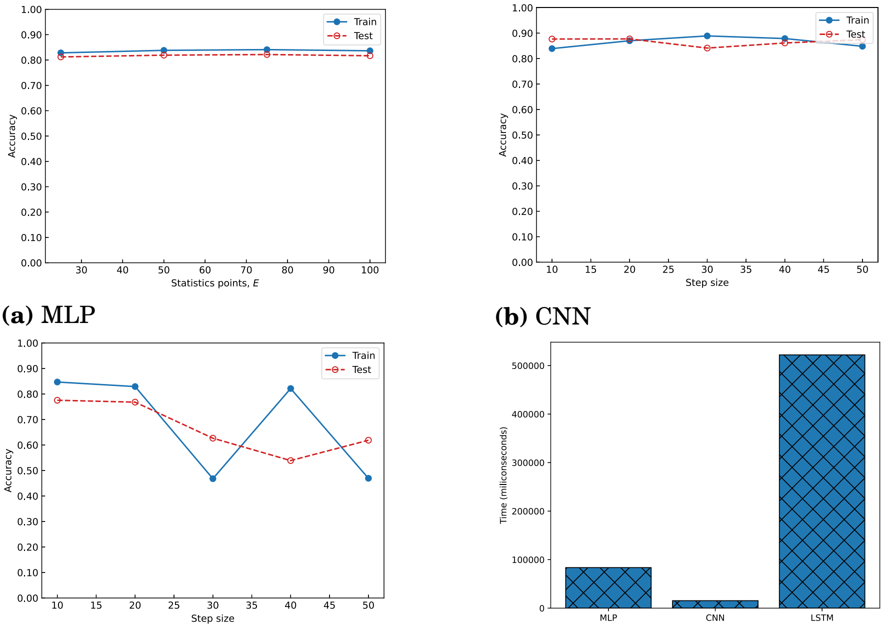
Figure 6.4. Kriging-based radio map construction

selected according to Wasserstein distance, and conspicuously a smaller Wasserstein distance corresponds to a higher possibility. After selection, the resampling uses the selected positions to compute new $\mu_{\varphi}, \kappa_{\varphi}$ and update the sampling PDF to make more particles toward the final position, as can be seen in Figure 6.3b and 6.3d. The algorithm of the radio map-based GPS spoofing mitigation method is detailed in Publication VII.

6.3 Performance Evaluation

In this section, we provide an analysis of the kriging-based 3D radio map construction method and an evaluation of the proposed particle filter-based GPS spoofing mitigation approach. To this end, we first use Blender 3.3 and radio gym to build an urban area and construct a 3D radio map [106], and then develop a GPS spoofing detection and mitigation simulator using Python 3.8.10 and Tensorflow 2.9.0. Specifically, the 3D radio map is in a 3D $300 \times 300 \times 50$ m^3 space, where a BS is set atop a build and use 2.5 GHz bandwidth for communication. The parameters of ray tracing and the deterministic model are summarized in Publication VII. Additionally, the particle filter has 200 sampling particles and 10 particles with the smallest Wasserstein distance being used for the PDF resampling. The simulation results are presented in what follows.

Figure 6.4 demonstrates Kriging-based radio map construction. Figure 6.4a is the baseline radio map where all of the points are measured with



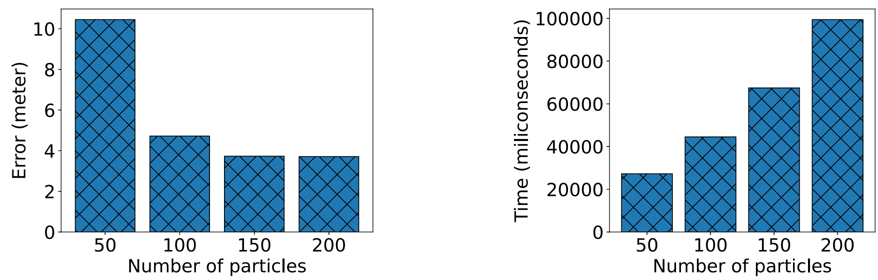
(a) MLP **(b)** CNN **(c)** LSTM **(d)** Train time

Figure 6.5. Radio map-based based GPS spoofing detection

ray tracing and a deterministic channel model. To reduce computation consumption, the Universal Kriging (UK) along with an exponential kernel is adopted to interpolate the sampling points, as illustrated in Figure 6.4b and Figure 6.4c, respectively. The UK interpolated radio map is close to the baseline map, which means the use of UK interpolation can maintain the radio map performance as well as reduce computation consumption. Figure 6.4d shows the final UK fine-grained 3D radio map used for GPS spoofing detection and mitigation, where the fine-grained radio map has a higher resolution on the UAV movement region and further improves the accuracy of the theoretical RSS values.

Figure 6.5 presents the efficiency of radio map-based spoofing detection methods, namely the MLP, the CNN, and the LSTM. Among these, the CNN stands out due to its superior accuracy in spoofing detection, achieved by its capability to unearth complex spoofing patterns from the path loss sequence. On the other hand, the LSTM, being a type of the RNN, tends to require more extensive time for training, which makes it more challenging to utilize. Considering accuracy and training time efficiency, the CNN is generally favored for spoofing detection.

Figure 6.6 depicts GPS spoofing mitigation performance, which is influenced by the number of particles. As shown in Figure 6.6a, the mitigation error is decreasing with increasing particles until a certain level, which is caused by inevitable errors in sampling and resampling. Conversely,



(a) GPS spoofing mitigation error

(b) GPS spoofing mitigation time

Figure 6.6. GPS spoofing mitigation performance

in Figure 6.6b, the mitigation time keeps the same trend as the number of particles because the complexity of mitigation is $O(N^2)$ that has been proven in Publication VII. Therefore, rather than replace GPS, the proposed mitigation method is designed for emergency usage after spoofing detection because of its limited accuracy and high latency.

6.4 Summary and Discussions

This chapter investigated 3D radio map-based GPS spoofing detection and mitigation methods for cellular-connected UAVs. The combination of ray tracing and deterministic channel models is the basis for constructing a 3D radio map, while ray tracing requires a large amount of computation on the edge server. To mitigate this, the UK interpolation method is adopted to reduce the amount of ray tracing required. Then, machine learning is introduced to detect and mitigate the GPS spoofing using a 3D radio map, where the MLP, CNN, and RNN are for GPS spoofing detection while a particle filter is used for the mitigation of GPS spoofing, respectively. The results indicate that the CNN performs better in terms of GPS spoofing detection while efficiently utilizing training resources, and the particle filter relocates the UAV with an error margin of 10 meters and mitigates GPS spoofing attacks.

Despite the achievements introduced by the radio map regarding GPS spoofing detection and mitigation, it is a challenge to build a 3D radio map in a big city because of the limited computation and storage resources. Compared with the state-of-the-art radio map construction methods, the proposed fine-grained radio map construction algorithms could build a radio map in a reasonable time while maintaining an acceptable level of accuracy. Although the proposed fine-grained radio map can help the UAV detect and mitigate GPS spoofing attacks, it still takes considerable resources and time to update the radio map in a highly dynamic UAV environment. In the future, we will elaborate more on federated learning-based 3D radio map construction. The federated learning approach could build a radio map that considers both radio measurement data and the radio propagation model and provide seamless GPS spoofing detection and mitigation for cellular-connected UAVs.

7. Conclusions

This dissertation presented the author's work on detecting and mitigating GPS spoofing attacks for cellular-connected UAVs. Despite cellular-connected UAVs delivering a wide range of applications, they are vulnerable to GPS spoofing attacks that can compromise the control and management functions of such vehicles. Indeed, GPS spoofing attacks can render UAVs uncontrollable making them violate airspace regulations. Although existing methods can help UAVs against GPS spoofing, these methods consume extra energy and require special hardware, or even are impractical due to environmental constraints. Thereby, this dissertation aims to develop comprehensive spoofing detection and mitigation methods for cellular-connected UAVs.

7.1 Discussion

In order to address the objective established by the dissertation, we investigate the 3GPP MPS that can provide identifying and tracing services to the UAV traffic management system, UTM. However, MPS technology typically has limitations in terms of coverage area and accuracy. The conventional MPS uses a triangulation algorithm to locate the UAV, which requires that the UAV be in an area covered by at least three BSs at the same time; otherwise, the MPS only certifies the UAV's position inside its coverage area or outside its coverage area. To this end, we have developed four research questions to investigate machine learning approaches for GPS spoofing detection and mitigation.

The first research question focused on improving MPS-based GPS spoofing detection accuracy under three base stations. In response to the MPS's low accuracy and the UAV's high dynamic, we have proposed an ATRA solution that can use 5G network facilities to monitor UAV GPS position. The ATRA leverages the uplink RSS measurements collected by the nearby BSs to compute the residence area, and that area is then divided into small parts and assigned to different trust levels. The evaluation of the ATRA shows the high effectiveness of GPS spoofing detection in both free-space and urban areas. However, the ATRA still requires at least three BSs at the same time, which limits its application in situations where the UAV is located inside the cell.

The second research question elaborated on using advanced deep neural networks to monitor the UAV positions and detect GPS spoofing in the area covered by multi-BSs or even a single BS. To detect GPS spoofing in the area covered by a single BS, we designed an MLP neural network that uses different kinds of path loss statics as inputs and outputs the spoofing possibility for UAV-reported GPS locations. Regardless of the performance brought by the MLP, its performance heavily depends on the amount of training data: the more data, the greater the accuracy of the model. In addition, the MLP uses path loss statics which may drop some original features from the raw data and hinder the spoofing behaviors. To overcome the weakness of the MLP, we designed a new CNN that can analyze the raw path loss data directly. However, the CNN is a deep neural network that spends a large amount of time on model training and updating, which may introduce detection latency in a highly dynamic UAV environment. To this end, transfer learning is introduced that can train a new CNN model based on the pre-trained model shared by another BS. The evaluation of transfer learning shows that the model training time is reduced and the model accuracy is increased. With regard to multi-BS scenarios, deep ensemble learning is used to combine the individual MLP results, which can further improve the deep neural networks' performance in terms of GPS spoofing detection accuracy. While deep ensemble learning is a valid approach, it needs to run the detection model for each UAV independently and takes too much computation resources when there is a flock of UAVs.

The third research question targeted delivering an efficient GPS spoofing detection service for cellular-connected UAV swarms. Indeed, a cellular-connected UAV swarm is more economical and efficient in different kinds of tasks. In addition, to save communication bandwidths, the UAV swarm is flocked as a FANET where only the swarm leader has a direct wireless connection with the BS. Due to the FANET's lesser security consideration for each node, it is easy for a spoofer to spoof the swarm GPS and change the swarm topology. Luckily, this can be detected by a GNN which can compute the spatial similarity between the swarm GPS topology and communications topology. In addition to spatial similarity, the temporal similarity changes are also important to indicate the GPS spoofing attacks. So, we designed a GNN that consists of GCNs layers and recurrent network layers. The evaluations show the proposed GNN is able to detect GPS spoofing attacks in a very short time, which is in the order of milliseconds.

The last research question concentrated on recovering the real position and mitigating GPS spoofing attacks for cellular-connected UAVs. Although the work in [59] allows the recovery of the real GPS signal from the spoofed ones, it can be further improved with the radio map and machine learning approaches demonstrated in this dissertation. To this end, we build a 3D radio map using ray tracing and deterministic channel models. Considering the fact that ray tracing is a heavy program, we use the UK interpolation method with an exponential kernel to construct the fine-grained 3D radio map. With the help of a 3D radio map, the particle filter is applied to relocate the UAV position after the spoofing detection. The evaluation results show that the particle filter is able to find a real position with an accuracy of within 10 meters.

7.2 Open Issues and Research Challenges

This dissertation addressed enhancing the performance of cellular-connected UAVs and designed a variety of neural networks for the edge UFC to detect and mitigate GPS spoofing attacks. Although a number of results demonstrated the efficiency of the proposed neural networks, there are the following topics that need to be investigated further.

Most of the proposed solutions in this dissertation are still in a quite nascent stage in terms of practical deployment. In the real world, the execution of the developed solutions needs collaboration from different stakeholders, such as mobile operators, standard organizations, and UTM. For instance, the proposed solutions require information captured from BSs (e.g., the position of BSs, the frequency of sub-carries, and the transmission power of BSs, etc.), and information from the UAV being investigated (e.g., the GPS position of UAV and the employed transmission power, etc.), as well as information from the UTM (e.g., the flight regulation and the authorized flight path, etc.), and the exact required information, depends on the tackled problem and the UAV flight scenario. In addition, the proposed solutions consider the BS equipped with an omnidirectional antenna. However, the 5G-and-beyond networks have already started the technologies such as MIMO antennas that are able to locate the ground user within decimeters. Therefore, it is possible to apply these new techniques in the air for cellular-connected UAVs. Furthermore, the proposed solutions are designed only for the scenario in which UAVs have connections with BSs. Nevertheless, other scenarios, such as the UAV in areas without cellular connections, need assistance from INS and other localization methods for spoofing detection and mitigation. This raises further research questions on the combination of different GPS spoofing detection and mitigation methods.

In terms of ML-based GPS spoofing detection approaches in this thesis, they are trained and tested with a centralized approach on an edge server. Exploring distributed approaches for spoofing detection and mitigation is another open research direction of the thesis. For instance, the GNN-based spoofing detection proposed in Chapter 5 is based on a centralized agent. The GNN collects the data from different UAVs, trains a model to compare the swarm GPS topology and communication topology, and decides whether the reported GPS position is spoofed or not. Such an approach leads to a decision delay because it needs to wait until it has received information from all of the swarm members. In addition, the UAV may not share its information due to privacy considerations. Therefore, it is very important to explore distributed approaches, such as federated learning-based GPS spoofing detection for cellular-connected UAV swarms.

In terms of the ML-based GPS spoofing mitigation approach in this thesis, it recovers the UAV's real position based on a 3D radio map for a mobile cellular network. Indeed, there is abundant radio frequency in the air, such as public TV signals and broadcast signals, which also can be used to build radio fingerprints in the air for the UAV to locate itself and mitigate GPS spoofing. However, it takes too much time and storage to build and save such a huge 3D radio map data. Thus, it is a new open research direction that exploits the radio fingerprint of air signals to mitigate the spoofing attack.

Another research direction that can be considered as a continuation of this work is non-GPS navigation for UAVs. Compared with 31 GPS satellites for navigation, the next generation space network, Starlink operated by SpaceX, already has 3,580 satellites in low earth orbit and nearly 12,000 satellites will be deployed, which can provide not only communication but also navigation for UAVs [107]. There are two kinds of approaches towards navigation with Starlink satellites, and one is based on a new navigation protocol while the other exploits a Kalman filter for analyzing satellite signals for navigation [108]. However, the navigation protocol consumes extra bandwidth while the signal analyzation is vulnerable to space noise. The solution proposed in this dissertation constitutes a basis that can be used by satellite-connected UAVs to realize safe and cheap Starlink navigation.

Indeed, while the proposed machine learning models have shown promising results in simulations, real-world validation is an open issue that requires thorough field experiments. The models should be tested in an environment that includes multiple UAVs operating simultaneously and across various carrier frequencies. It is essential to truly assess the efficacy of these models under real-world conditions. Consequently, integrating these models into an Open-RAN system is a critical future step, offering a promising approach to counter GPS spoofing attacks for cellular-connected UAVs, thereby ensuring increased security and reliability for UAV operations.

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